MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE EBL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.099 failures per million operating hours for class I or 0.272 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/41-1 for class I and M28787/41-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

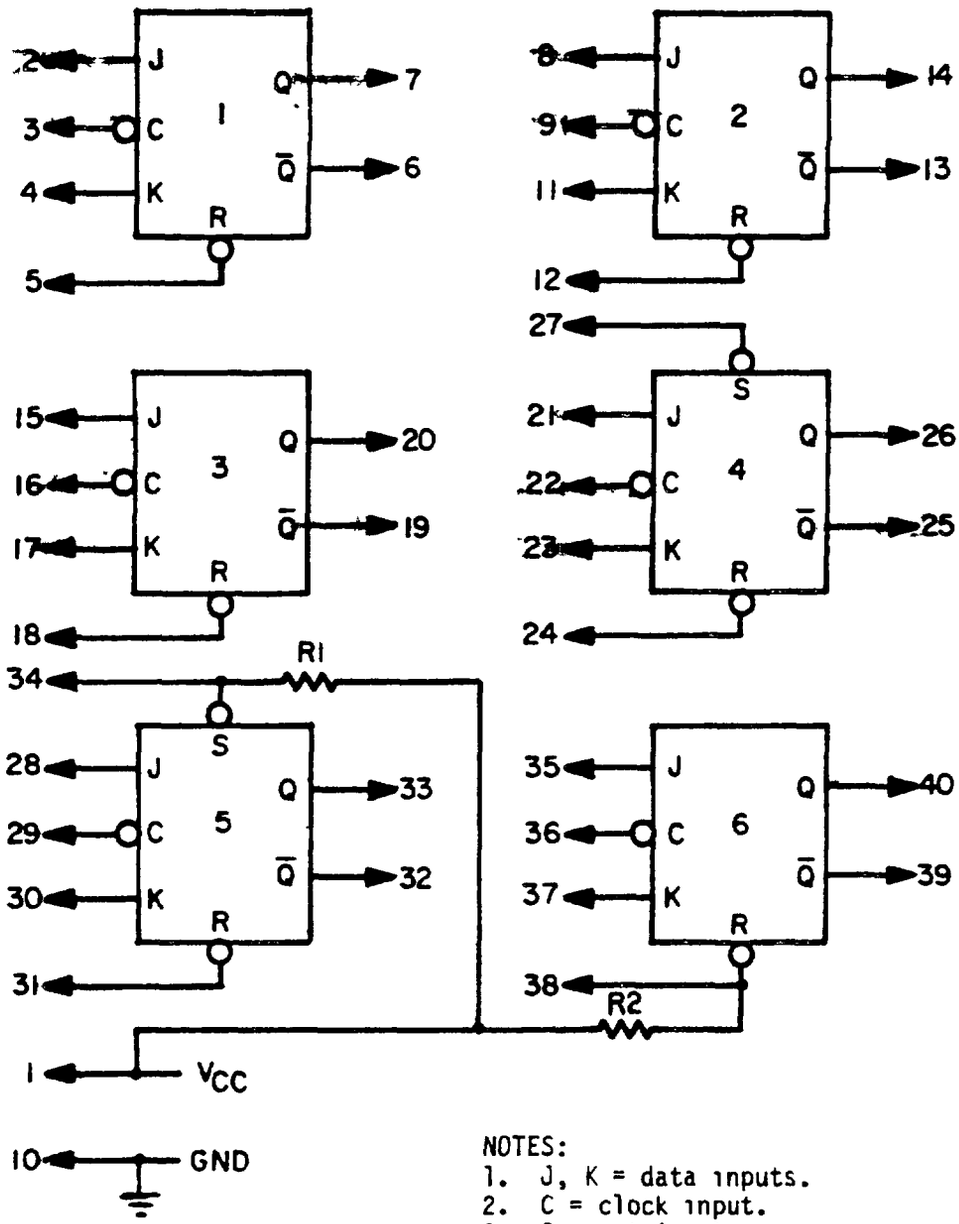
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES:
1. J, K = data inputs.
2. C = clock input.
3. S = set input.
4. R = reset input.
5. Q, $\overline{Q}$ = outputs.
6. V_{CC} = 5.0 V dc ±10%.
7. GND = 0 V REF.
8. R1 = 10 kΩ.
9. R2 = 4.7 kΩ.

FIGURE 1. EBL module function diagram, JK flip-flop.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C26	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D12	Output short circuit current	VI	X	-20	-57	-20	-57	-20	-57	-20	-57	-20	-57	mA
E1-E26	Input clamp diode voltage	VII	S	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
F1-F16	Functional	VIII	X	← See table VIII →										
G1-G12	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H12	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J12	Input current, high level	X	X	0	40	0	40	0	40	0	40	0	40	µA
J13-J18	Input current, high level	X	X	-850	80	-850	80	-850	80	-850	80	-850	80	µA
J19-J24	Input current, high level	X	X	0	80	0	80	0	80	0	80	0	80	µA
J25	Input current, high level	X	X	-210	-320	-210	-320	-210	-320	-210	-320	-210	-320	µA
J26	Input current, high level	X	X	-540	-690	-540	-690	-540	-690	-540	-690	-540	-690	µA
K1-K12	Input current, low level	XI	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
K13-K24	Input current, low level	XI	X	-0.20	-6.92	-0.20	-6.92	-0.20	-6.92	-0.20	-6.92	-0.20	-6.92	mA
K25	Input current, low level	XI	X	-0.6	-7.5	-0.6	-7.5	-0.6	-7.5	-0.6	-7.5	-0.6	-7.5	mA
K26	Input current, low level	XI	X	-1.20	-8.15	-1.20	-8.15	-1.20	-8.15	-1.20	-8.15	-1.20	-8.15	mA
L1,L2	Power supply current	XII	X	40	120	40	120	40	120	40	120	40	120	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M12	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	10	45	10	50	10	50	10	50	10	50	ns
M13-M24	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	10	55	10	60	10	60	10	60	10	60	ns
N1-N12	Transition time (t_{THL}), C_L = 50 pF	XIII	S	5	25	5	30	5	30	5	30	5	30	ns
N13-N24	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	5	35	5	40	5	40	5	40	5	40	ns
P1-P12	Propagation delay time (t_{PLH}), C_L = 50 pF	XIII	S	10	30	10	35	10	35	10	35	10	35	ns
P13-P24	Propagation delay time (t_{PLH}), C_L = 150 pF	XIII	Q	10	39	10	44	10	44	10	44	10	44	ns
Q1-Q12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIII	S	10	44	10	49	10	49	10	49	10	49	ns
Q13-Q24	Propagation delay time (t_{PHL}), C_L = 150 pF	XIII	Q	10	54	10	59	10	59	10	59	10	59	ns
R1,R2, S1-S6	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	5	29	5	34	5	34	5	34	5	34	ns
R3,R4, S7-S12	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	5	39	5	44	5	44	5	44	5	44	ns
T1,T2, U1-U6	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	10	44	10	49	10	49	10	49	10	49	ns
T3,T4, U7-U12	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	10	54	10	59	10	59	10	59	10	59	ns
V1-V6	Maximum clock frequency check, C_L = 50 pF	XV	S	200	212	200	212	200	212	200	212	200	212	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W1-W12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
X1-X12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply filter capacitance	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
AA1-AA12	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
AA13-AA26	Input capacitance	XIX	Q	2	24	2	24	2	24	2	24	2	24	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Part number M28787/41-1 supersedes part number M28787/41. The two parts are interchangeable.

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* 6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation	660 max	660 max	mW
Supply voltage	5 ±10%	5 ±10%	V dc
Input voltage			
J, K, C1-C6, R1-R5, S4			
Low level	-0.5 min,	-0.5 min,	V
	0.8 max	0.8 max	V
High level	2 min,	2 min,	V
	V_{CC} max	V_{CC} max	V
R6, S5			
Low level	-0.5 min,	-0.5 min,	V
	0.8 max	0.8 max	V
Input current			
C1-C6			
Low level (V_{IL} = 0.0 V)	-0.2 min, -6.92 max	-0.2 min, -6.92 max	mA
(V_{IL} = 0.4 V)	--- -3.20 max	--- -3.20 max	mA
High level (V_{IH} = 2.4 V)	-850 min, 80 max	-850 min, 80 max	µA
(V_{IH} = 5.5 V)	--- 1 max	--- 1 max	mA
R6			
Low level (V_{IL} = 0.0 V)	-1.20 min, -8.15 max	-1.20 min, -8.15 max	mA
(V_{IL} = 0.4 V)	-1.20 min, -4.34 max	-1.20 min, -4.34 max	mA
High level (V_{IH} = 2.4 V)	-0.54 min, -0.69 max	-0.54 min, -0.69 max	mA
(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
J, K			
Low level (V_{IL} = 0.0 V)	-0.10 min, -1.73 max	-0.10 min, -1.73 max	mA
(V_{IL} = 0.4 V)	-0.1 min, -1.6 max	-0.1 min, -1.6 max	mA
High level (V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
R1-R5, S4			
Low level (V_{IL} = 0.0 V)	-0.20 min, -6.92 max	-0.20 min, -6.92 max	mA
(V_{IL} = 0.4 V)	-0.2 min, -3.2 max	-0.2 min, -3.2 max	mA
High level (V_{IH} = 2.4 V)	0 min, 80 max	0 min, 80 max	µA
(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
S5			
Low level (V_{IL} = 0.0 V)	-0.6 min, -7.5 max	-0.6 min, -7.5 max	mA
(V_{IL} = 0.4 V)	-0.60 min, -3.74 max	-0.60 min, -3.74 max	mA
High level (V_{IH} = 2.4 V)	-0.21 min, -0.32 max	-0.21 min, -0.32 max	mA
(V_{IH} = 5.5 V)	-0.21 min, 1.00 max	-0.21 min, 1.00 max	mA
Output voltage			
Low level (I_{OL} = 16 mA)	0.0 min, 0.4 max	0.0 min, 0.4 max	V
High level (I_{OH} = -400 µA)	2.4 min, 4.5 max	2.4 min, 4.5 max	V
Output drive current, maximum capability			
Low level (V_{OL} = 0.4 V)	--- 16 max	--- 16 max	mA
High level (V_{OH} = 2.4 V)	--- -400 max	--- -400 max	µA
Clock input			
Frequency	-- 10 max	10 max	MHz
Pulse width (high level)	20 min	20 min	ns
Set or reset pulse width (low level)	25 min	25 min	ns

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Input setup time	≥ clock pulse width	≥ clock pulse width	
Input hold time	0.0 min	0.0 min	ns

* 6.6 Module function. This module contains six JK flip-flops with both true (Q) and complementary ($\overline{Q}$) outputs as shown in figure 1. Each JK flip-flop has independent J, K, clock, and reset inputs. Circuits 4 and 5 have the set input also available. Data is transferred into a temporary memory when the clock goes high. When the clock returns low, the data is transferred to the Q and $\overline{Q}$ outputs. The data on the J and K inputs should be present when the clock goes high and should not be changed while the clock is high. The application of a low level to the reset input will cause the Q output to go low. The application of a low state to the set input will cause the Q output to go to a high state. The set and reset inputs are independent of the clock. Pull-up resistors are available on the S5 and R6 inputs with sufficient capability for pulling up the several other inputs.

6.6.1 Input to output relationship. Each JK flip-flop has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 7/

INPUTS					OUTPUTS			
Set	Reset	J	K	Clock	Qn	Qn+ 1	$\overline{Q}$n	$\overline{Q}$n+ 1
H	L	X	X	X	L	L	H	H
L	H	X	X	X	H	H	L	L
H	H	L	L	Pulse	L	L	H	H
H	H	L	L	Pulse	H	H	L	L
H	H	L	H	Pulse	L	L	H	H
H	H	L	H	Pulse	H	L	L	H
H	H	H	L	Pulse	L	H	H	L
H	H	H	L	Pulse	H	H	L	L
H	H	H	H	Pulse	L	H	H	L
H	H	H	H	Pulse	H	L	L	H

1/ H = High level.

2/ L = Low level.

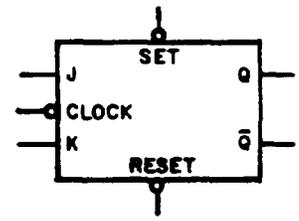
3/ X = Don't care.

4/ Qn = Q output before clock pulse.

5/ Qn+1 = Q output after clock pulse.

6/ $\overline{Q}$n = $\overline{Q}$ output before clock pulse.

7/ $\overline{Q}$n+1 = $\overline{Q}$ output after clock pulse.



6.6.2 Parallel outputs. Paralleling of outputs is not permissible.

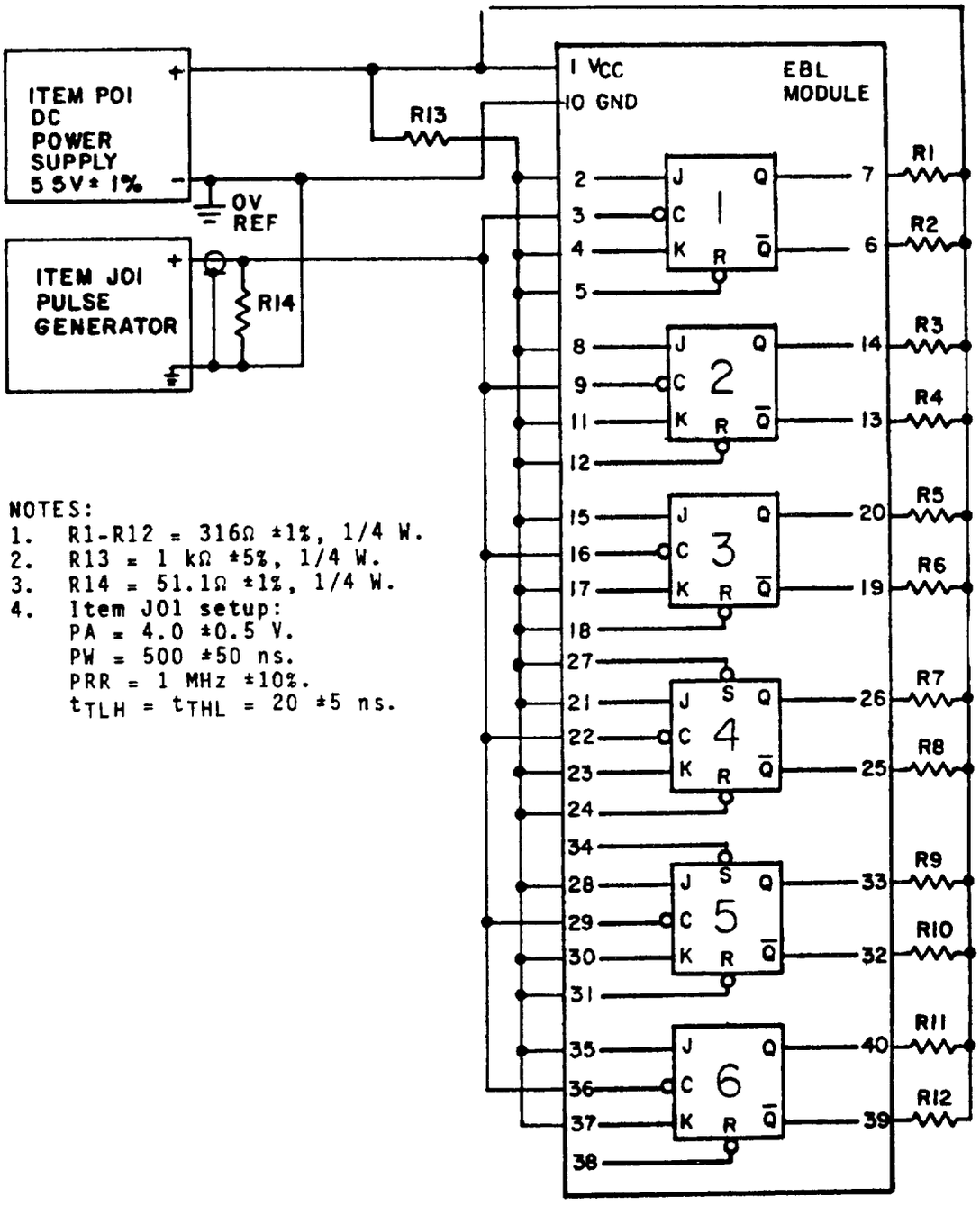
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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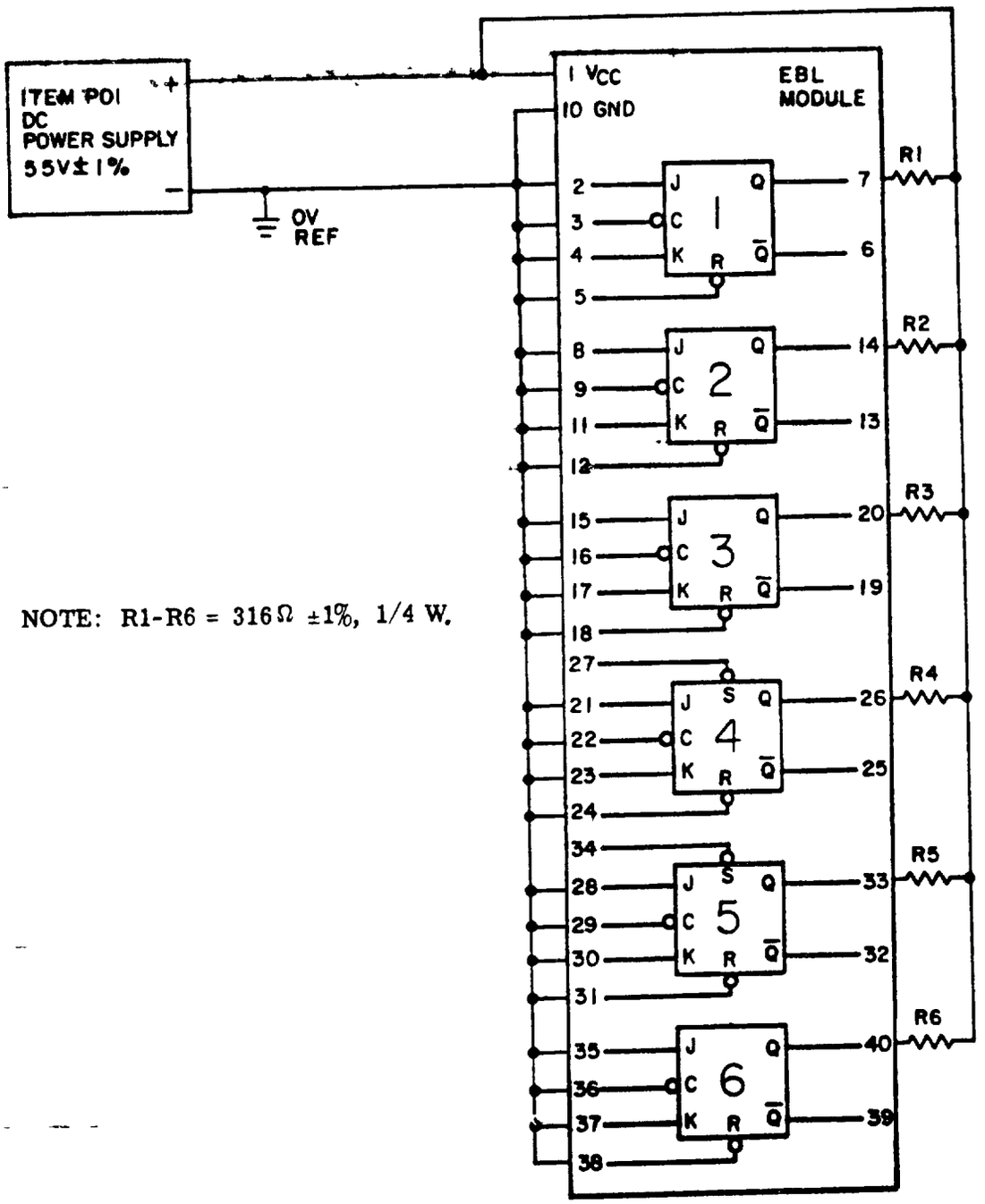
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NOTES:
1. R1-R12 = 316Ω ±1%, 1/4 W.
2. R13 = 1 kΩ ±5%, 1/4 W.
3. R14 = 51.1Ω ±1%, 1/4 W.
4. Item J01 setup:
   PA = 4.0 ±0.5 V.
   PW = 500 ±50 ns.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL} = 20 \pm 5$ ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R6 = 316Ω ±1%, 1/4 W.

* FIGURE 3. Component temperature test circuit.

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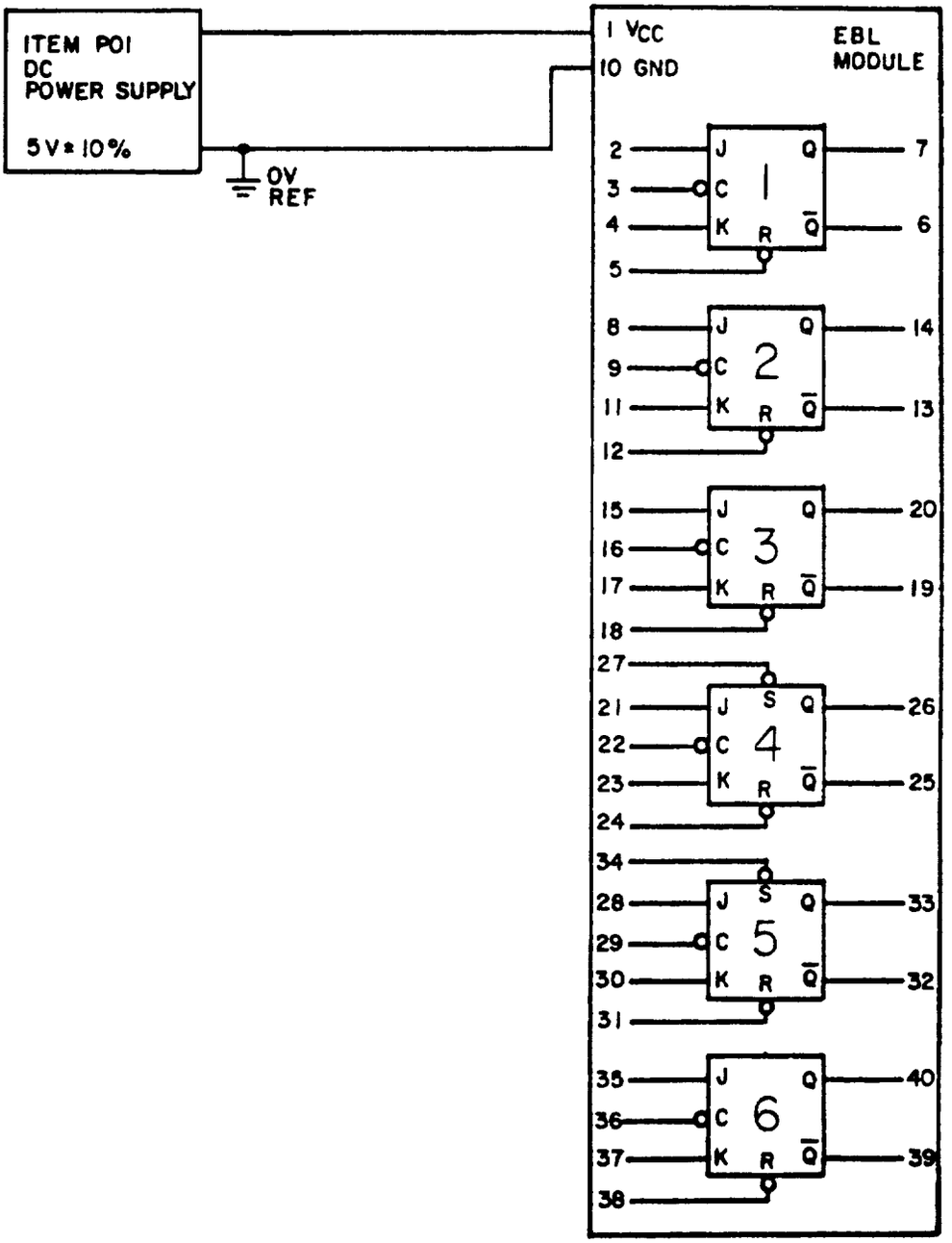


FIGURE 4. Vibration test circuit.

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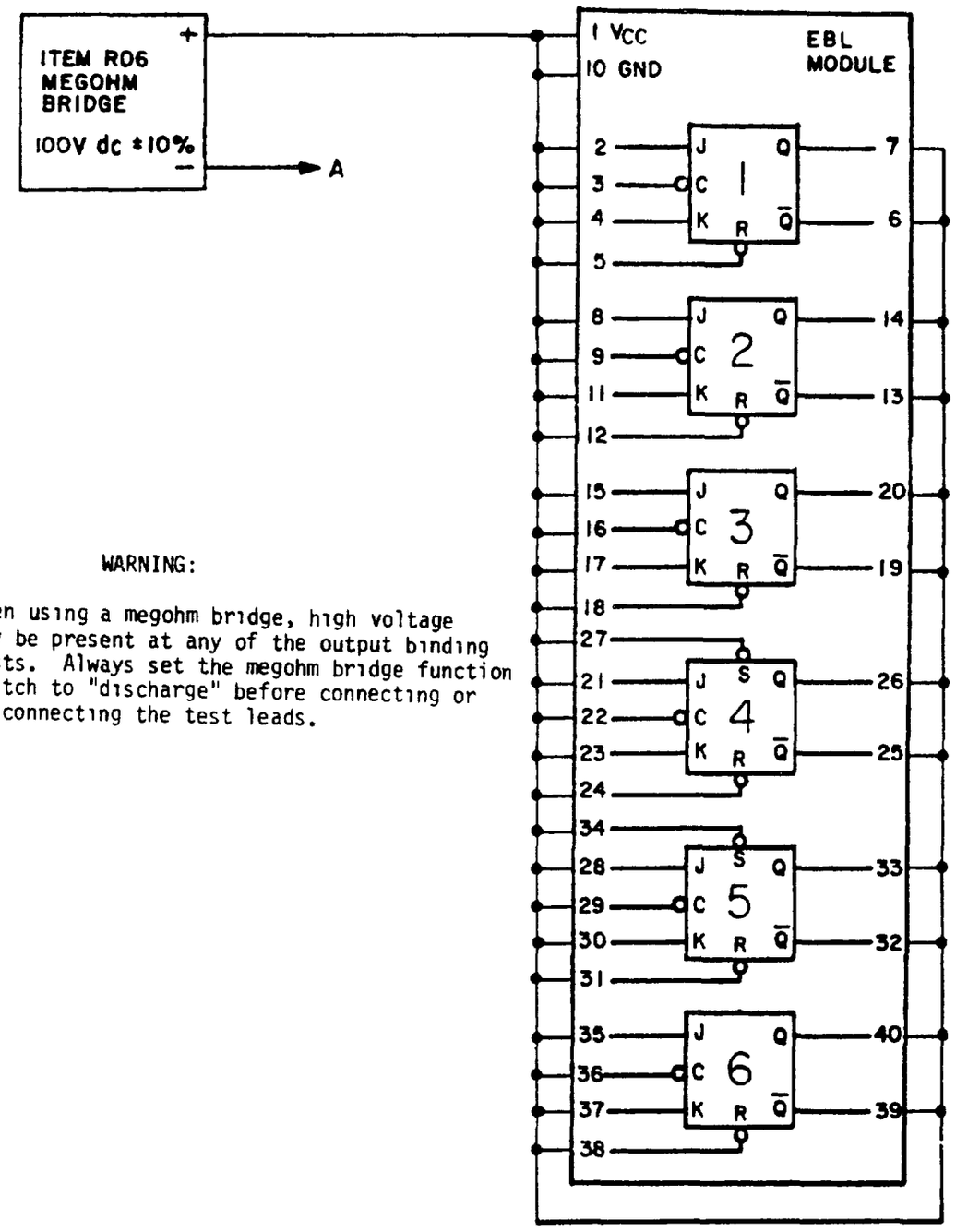
TABLE III. Isolation test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Negative to pin	Min	Max		
A	Isolation	Frame	10	∞	MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	1Q	V_{OUT} (low loaded)	7	Open	5	10	800	mV	6
B2	2Q	"	14	"	12	"	"	"	"
B3	3Q	"	20	"	18	"	"	"	"
B4	4Q	"	26	"	24	"	"	"	"
B5	5Q	"	33	"	31	"	"	"	"
B6	6Q	"	40	"	38	"	"	"	"
B7	$1\overline{Q}$	"	6	3	4	"	"	"	"
B8	$2\overline{Q}$	"	13	9	11	"	"	"	"
B9	$3\overline{Q}$	"	19	16	17	"	"	"	"
B10	$4\overline{Q}$	"	25	22	23	"	"	"	"
B11	$5\overline{Q}$	"	32	29	30	"	"	"	"
B12	$6\overline{Q}$	"	39	36	37	"	"	"	"

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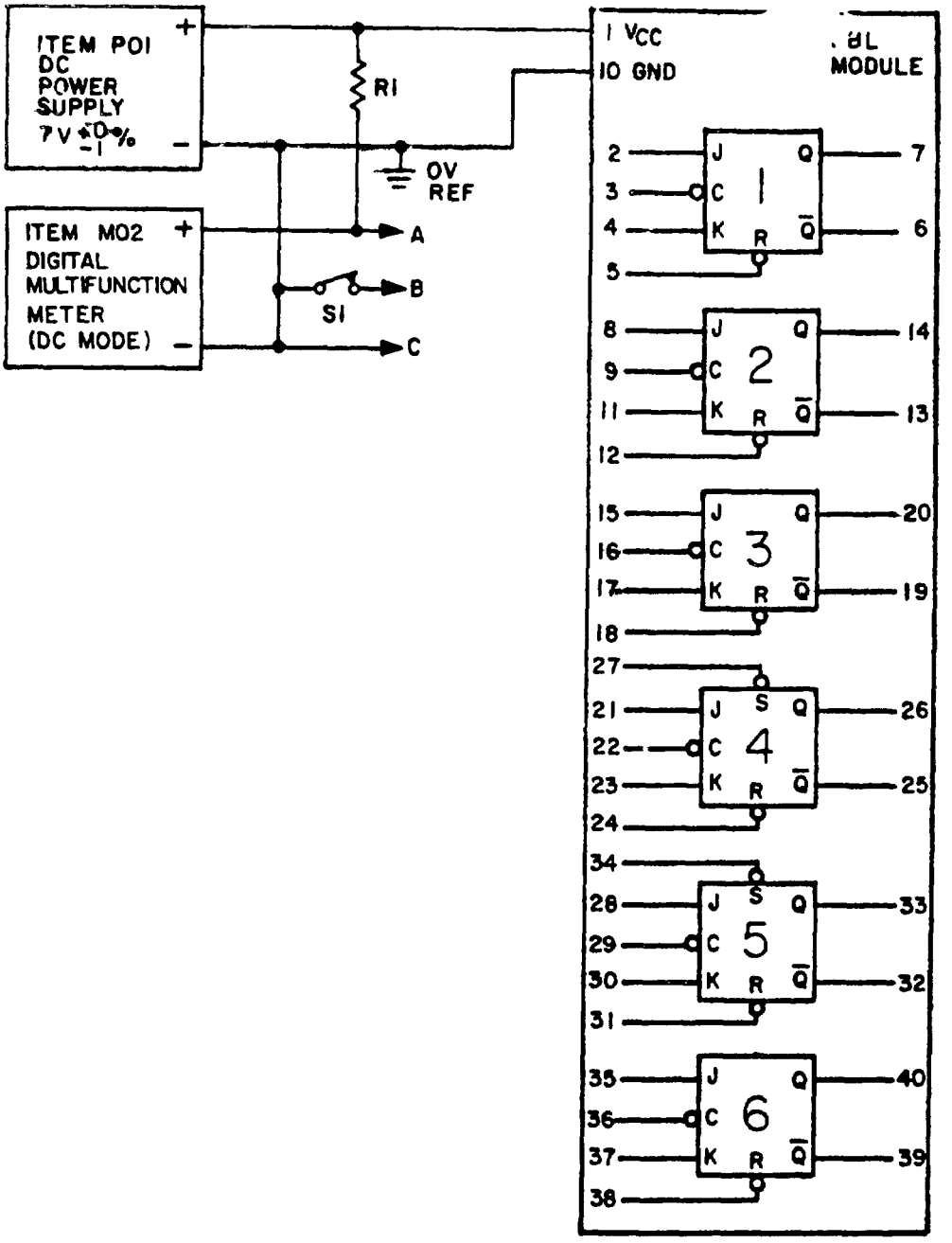
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table LI

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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. S1 = SPST switch.
3. Before connecting point A (S1 initially closed), open S1 for a minimum of 100 ns, then close it.
4. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1J Input breakdown current	3,4,5,8,9,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 34,35,36,37,38	2	0	1	mA	7
C2	1C "	2,4,5,8,9,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	3	"	"	"	"
C3	1K "	2,3,5,8,9,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	4	"	"	"	"
C4	1R "	2,3,4,8,9,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	5	"	"	"	"
C5	2J "	2,3,4,5,9,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	8	"	"	"	"
C6	2C "	2,3,4,5,8,11,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	9	"	"	"	"
C7	2K "	2,3,4,5,8,9,12,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	11	"	"	"	"
C8	2R "	2,3,4,5,8,9,11,15,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	12	"	"	"	"
C9	3J "	2,3,4,5,8,9,11,12,16,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	15	"	"	"	"
C10	3C "	2,3,4,5,8,9,11,12,15,17,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	16	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C11	3K	Input breakdown current	2,3,4,5,8,9,11,12,15,16,18, 21,22,23,24,27,28,29,30,31, 35,36,37,38,34	17	0	1	mA	7
C12	3R	"	2,3,4,5,8,9,11,12,15,16,17, 21,22,23,24,27,28,29,30,31, 34,35,36,37,38	18	"	"	"	"
C13	4S	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,28,29,30,31, 34,35,36,37,38	27	"	"	"	"
C14	4J	"	2,3,4,5,8,9,11,12,15,16,17, 18,22,23,24,28,29,30,31,34, 35,36,37,38	21	"	"	"	"
C15	4C	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,23,24,28,29,30,31,34, 35,36,37,38	22	"	"	"	"
C16	4K	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,24,28,29,30,31,34, 35,36,37,38	23	"	"	"	
C17	4R	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,28,29,30,31,34, 35,36,37,38	24	"	"	"	"
C18	5S	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 31,35,36,37,38	34	"	"	"	"
C19	5J	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,29,30,31, 35,36,37,38	28	"	"	"	"
C20	5C	"	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,30,31, 35,36,37,38	29	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C21	5K Input breakdown current	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,31, 35,36,37,38	30	0	1	mA	7
C22	5R "	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 35,36,37,38	31	"	"	"	"
C23	6J "	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 31,34,36,37,38	35	"	"	"	"
C24	6C "	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 31,34,35,37,38	36	"	"	"	"
C25	6K "	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 31,34,35,36,38	37	"	"	"	"
C26	6R "	2,3,4,5,8,9,11,12,15,16,17, 18,21,22,23,24,27,28,29,30, 31,34,35,36,37	38	"	"	"	"

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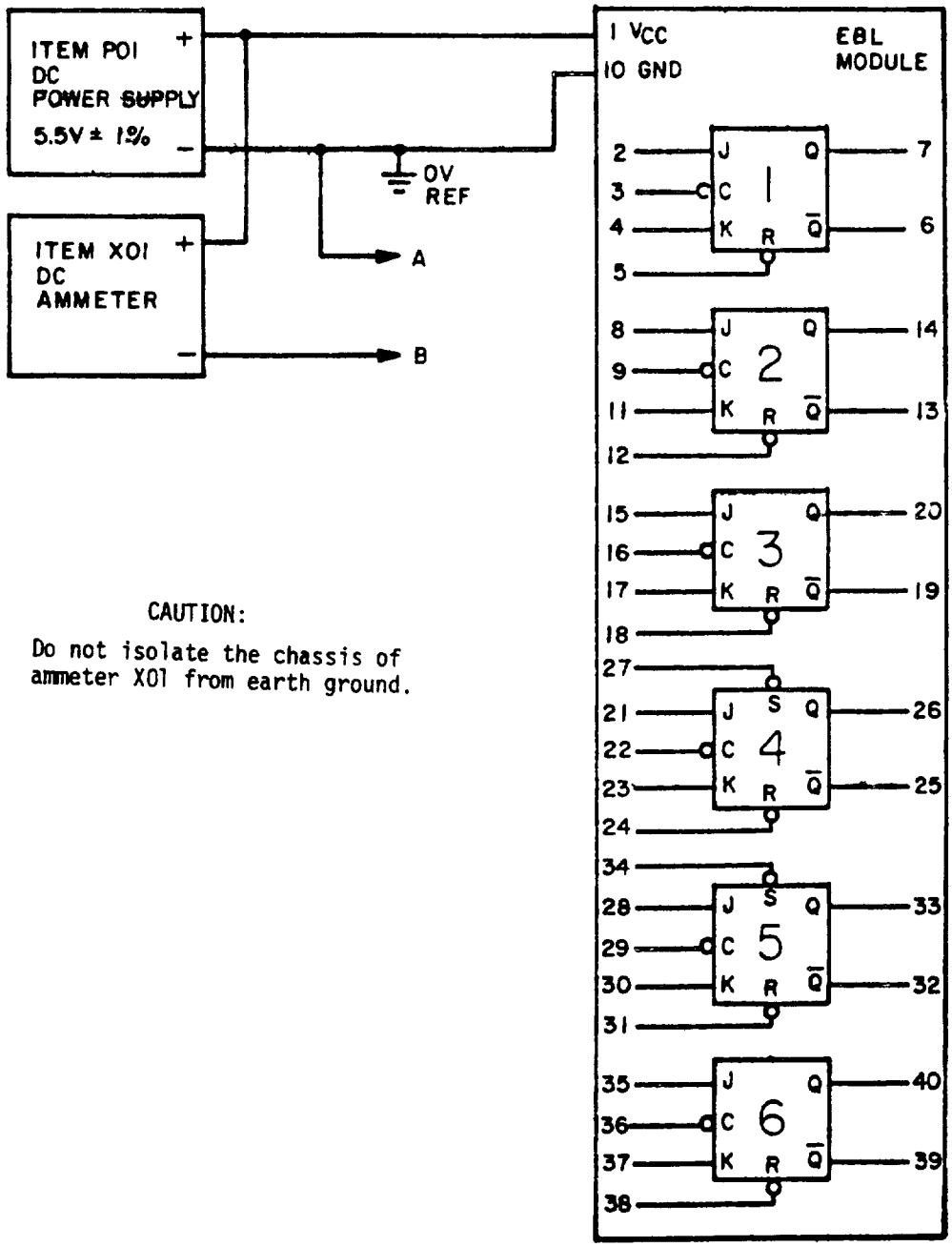


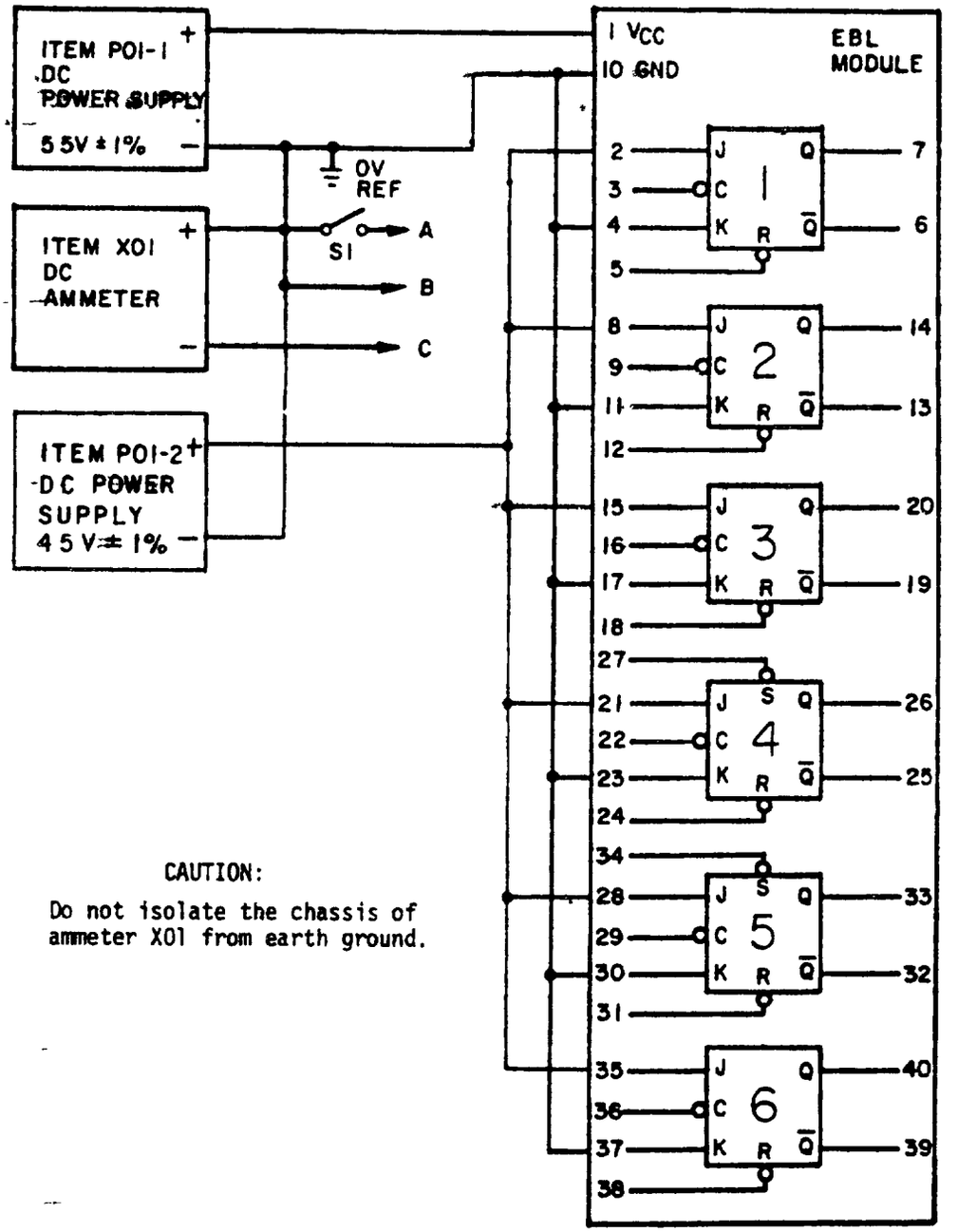
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1Q Output short circuit current	3	Open	7	-20	-57	mA	8
D2	2Q "	9	"	14	"	"	"	"
D3	3Q "	16	"	20	"	"	"	"
D4	4Q "	22	"	26	"	"	"	"
D5	5Q "	29	"	33	"	"	"	"
D6	6Q "	36	"	40	"	"	"	"
D7	$1\overline{Q}$ "	Open	5	6	"	"	"	"
D8	$2\overline{Q}$ "	"	12	13	"	"	"	"
D9	$3\overline{Q}$ "	"	18	19	"	"	"	"
D10	$4\overline{Q}$ "	"	24	25	"	"	"	"
D11	$5\overline{Q}$ "	"	31	32	"	"	"	"
D12	$6\overline{Q}$ "	"	38	39	"	"	"	"

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NOTES:
1. S1 = SPST switch.
2. Before connecting point C (S1 initially closed), open S1 for a minimum of 100 ns, then close it.
3. Only one output shall be shorted at a time and the reading taken within 5 seconds.

FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

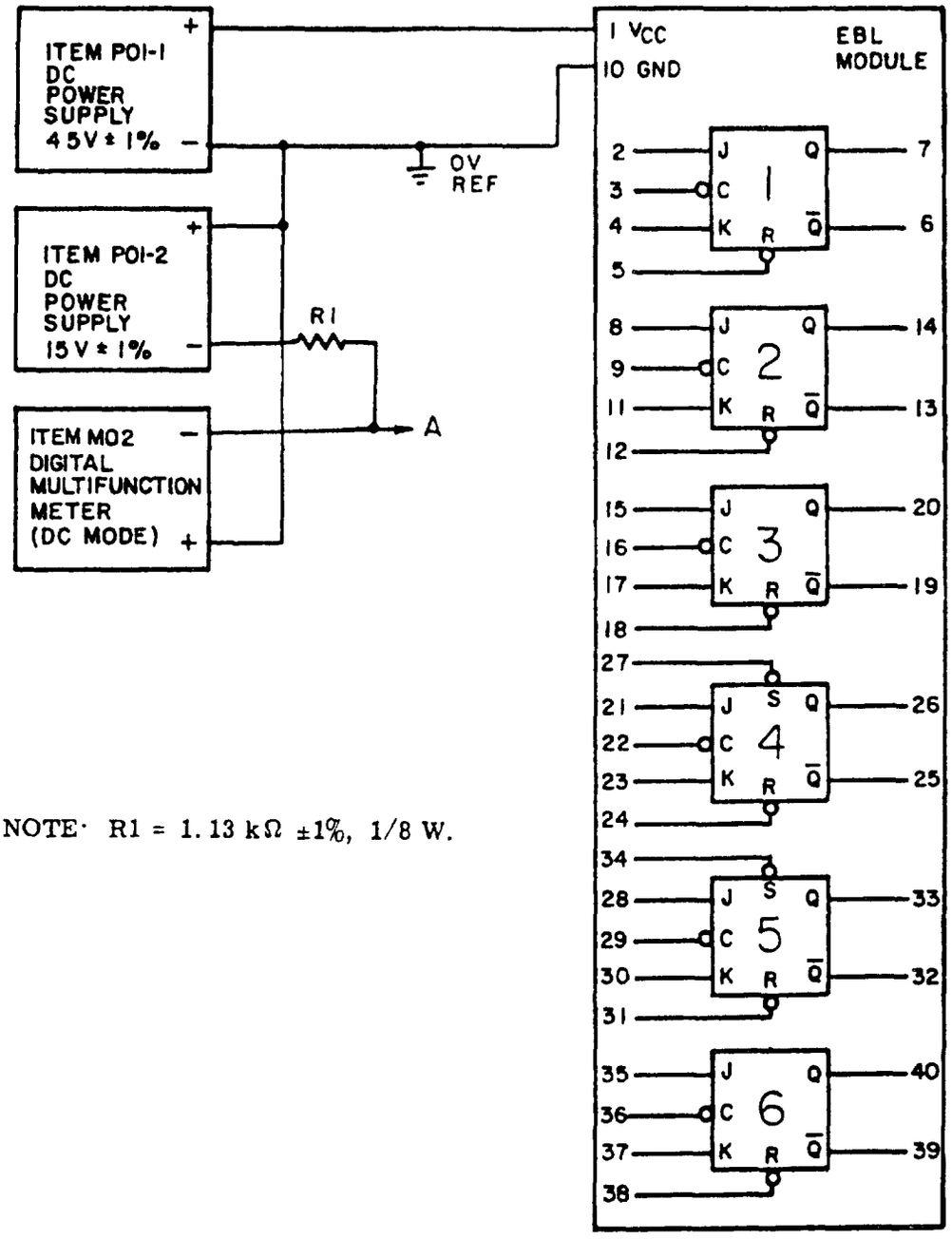
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E1	1J	Input clamp diode voltage	2	-0.2	-1.5	V	9
E2	1C	"	3	"	"	"	"
E3	1K	"	4	"	"	"	"
E4	1R	"	5	"	"	"	"
E5	2J	"	8	"	"	"	"
E6	2C	"	9	"	"	"	"
E7	2K	"	11	"	"	"	"
E8	2R	"	12	"	"	"	"
E9	3J	"	15	"	"	"	"
E10	3C	"	16	"	"	"	"
E11	3K	"	17	"	"	"	"
E12	3R	"	18	"	"	"	"
E13	4S	"	27	"	"	"	"
E14	4J	"	21	"	"	"	"
E15	4C	"	22	"	"	"	"
E16	4K	"	23	"	"	"	"
E17	4R	"	24	"	"	"	"
E18	5S	"	34	"	"	"	"
E19	5J	"	28	"	"	"	"
E20	5C	"	29	"	"	"	"
E21	5K	"	30	"	"	"	"
E22	5R	"	31	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E23	6J	Input clamp diode voltage	35	-0.2	-1.5	V	9
E24	6C	"	36	"	"	"	"
E25	6K	"	37	"	"	"	"
E26	6R	"	38	"	"	"	"

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Functional test.

Test condition 4/	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Output logic states 3/: 1Q	2Q	3Q	4Q	5Q	6Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	4$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	Initial limits 25°C ±5°C ambient: Min	Max	Unit	Test figure
F1	Functional, reset J = Low K = Low R = Low	27	2,4,5,8,11, 12,15,17, 18,21,23, 24,28,30, 31,35,37,38	2/	3,9,16, 22,29,36	L	L	L	L	L	L	H	H	H	H	H	H	1/			10
F2	Functional, reset J = Low K = High R = Low	4,11,17,23, 27,30,37	2,5,8,12, 15,18,21, 24,28,31, 35,38	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F3	Functional, reset J = High K = High R = Low	2,4,8,11, 15,17,21, 23,27,28, 30,35,37	5,12,18, 24,31,38	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F4	Functional, reset J = High K = Low R = Low	2,8,15,21, 27,28,35	4,5,11,12, 17,18,23, 24,30,31, 37,38	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F5	Functional, toggle J = High K = High R = High	2,4,5,8,11, 12,15,17, 18,21,23, 24,27,28, 30,31,35,37	Open	"	"	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F6	Functional, toggle J = High K = High R = High	2,4,5,8,11, 12,15,17, 18,21,23, 24,27,28, 30,31,35,37	Open	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F7	Functional, JK J = High K = Low R = High	2,5,8,12, 15,18,21, 24,27,28, 31,35	4,11,17,23, 30,37	"	"	H	H	H	H	H	H	L	L	L	L	L	L	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition 4/	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Output logic states 3/: 1Q	2Q	3Q	4Q	5Q	6Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	4$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	Initial limits 25°C ±5°C ambient: Min	Max	Unit	Test figure
F8	Functional, JK J = High K = Low R = High	2,5,8,12, 15,18,21, 24,27,28, 31,35	4,11,17,23, 30,37	2/	3,9,16, 22,29,36	H	H	H	H	H	H	L	L	L	L	L	L	1/			10
F9	Functional, no toggle J = Low K = Low R = High	5,12,18,24, 27,31	2,4,8,11, 15,17,21, 23,28,30, 35,37	"	"	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F10	Functional, JK J = Low K = High R = High	4,5,11,12, 17,18,23, 24,27,30, 31,37	2,8,15,21, 28,35	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F11	Functional, JK J = Low K = High R = High	4,5,11,12, 17,18,23, 24,27,30, 31,37	2,8,15,21, 28,35	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F12	Functional, no toggle J = Low K = Low	5,12,18,24, 27,31	2,4,8,11, 15,17,21, 23,28,30, 35,37	"	"	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F13	Functional, set J = Low K = Low	24,31	21,23,27, 28,30,34	"	22,29				H	H					L	L		"			"
F14	Functional, set J = High K = High	21,23,24, 28,30,31	27,34	"	"				H	H					L	L		"			"
F15	Functional, set J = High K = Low	21,24,28, 31	23,27,30,34	"	"				H	H					L	L		"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition 4/	Parameter	Connect				Output logic states 3/												Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	4$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	Min	Max		
F16	Functional, set J = Low K = High	23,24,30, 31	21,27,28, 34	2/	22,29				H	H					L	L		1/			10

1/ Output logic states: L = low level = 0.0 to 0.4 V; H = high level = 2.4 to 4.5 V.

2/ For test conditions F1-F16, verification of the output logic stated in footnote 1 is obtained by connecting item M02, point C, to each output pin in succession after input conditions have been accomplished.

3/ Q1, Q2, Q3, Q4, Q5, Q6, $\overline{Q}$1, $\overline{Q}$2, $\overline{Q}$3, $\overline{Q}$4, $\overline{Q}$5, and $\overline{Q}$6 output logic states are measured at output pins 7, 14, 20, 26, 33, 40, 6, 13, 19, 25, 32, and 39 respectively.

4/ Test conditions F1-F16 are to be run in sequence with only the applicable inputs to be changed.

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TABLE IX. Output voltage (low-level and high-level) test.

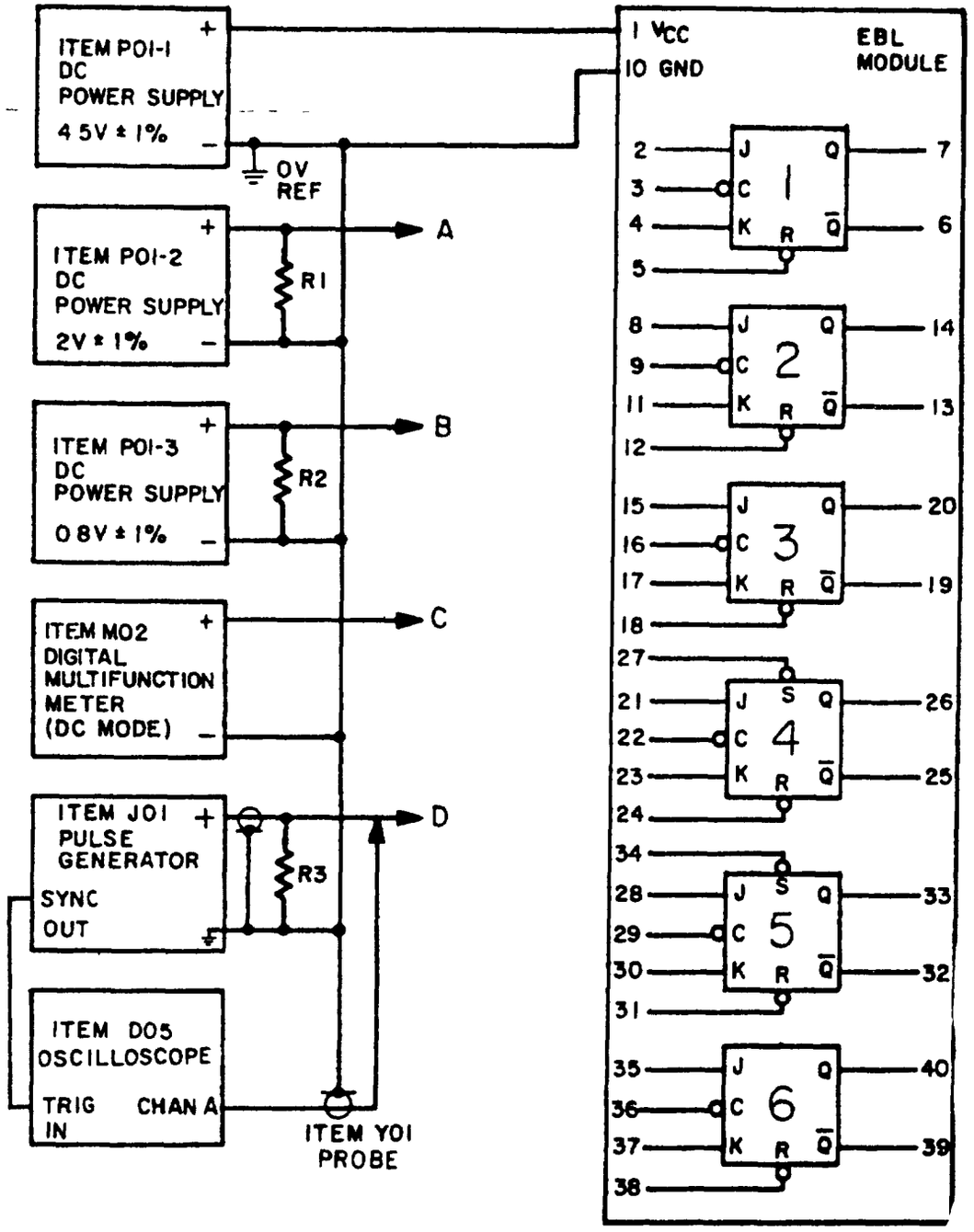
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q Output voltage, low level	4,5	2	7	3	0	400	mV	10
G2	2Q "	11,12	8	14	9	"	"	"	"
G3	3Q "	17,18	15	20	16	"	"	"	"
G4	4Q "	23,24, 27	21	26	22	"	"	"	"
G5	5Q "	30,31	28	33	29	"	"	"	"
G6	6Q "	37	35	40	36	"	"	"	"
G7	$1\overline{Q}$ "	2,5	4	6	3	"	"	"	"
G8	$2\overline{Q}$ "	8,12	11	13	9	"	"	"	"
G9	$3\overline{Q}$ "	15,18	17	19	16	"	"	"	"
G10	$4\overline{Q}$ "	21,24, 27	23	25	22	"	"	"	"
G11	$5\overline{Q}$ "	28,31	30	32	29	"	"	"	"
G12	$6\overline{Q}$ "	35	37	39	36	"	"	"	"
H1	1Q Output voltage, high level	2,5	4	7	3	2.4	4.5	V	"
H2	2Q "	8,12	11	14	9	"	"	"	"
H3	3Q "	15,18	17	20	16	"	"	"	"
H4	4Q "	21,24, 27	23	26	22	"	"	"	"
H5	5Q "	28,31	30	33	29	"	"	"	"
H6	6Q "	35	37	40	36	"	"	"	"

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TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H7	$1\overline{Q}$ Output voltage, high level	4,5	2	6	3	2.4	4.5	V	10
H8	$2\overline{Q}$ "	11,12	8	13	9	"	"	"	"
H9	$3\overline{Q}$ "	17,18	15	19	16	"	"	"	"
H10	$4\overline{Q}$ "	23,24, 27	21	25	22	"	"	"	"
H11	$5\overline{Q}$ "	30,31	28	32	29	"	"	"	"
H12	$6\overline{Q}$ "	37	35	39	36	"	"	"	"

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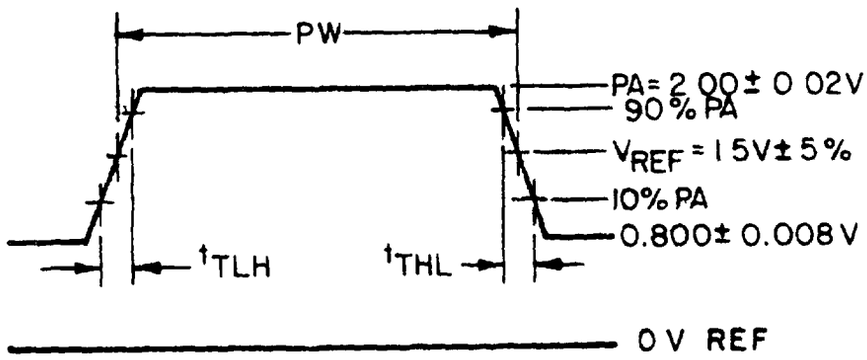


NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.
3. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 10. Functional and output voltage (low-level and high-level) test circuit for tables VIII and IX.

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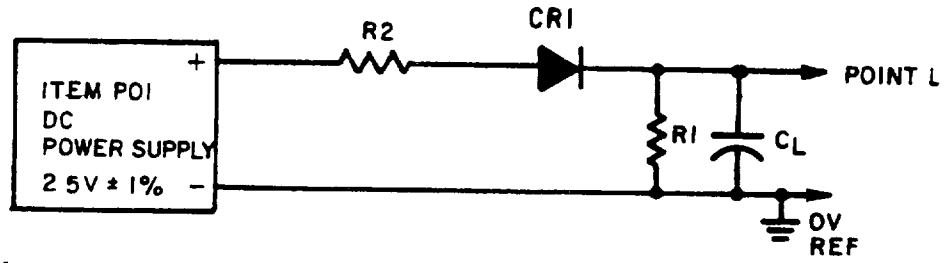
NOTES: - Continued
4. Load is not required for functional test F1-F16.
5. Item J01 setup:
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 25 \pm 5$ ns.
   PRR = 1 MHz ±10%.



The PRR is used for setup purposes only; when the pulse has been setup, the pulse generator should be placed in the one shot mode.
6. After all input conditions are setup per each test condition, apply one pulse using item J01. Then make the required measurements.

FIGURE 10. Functional and output voltage (low-level and high-level) test circuit for tables VIII and IX - Continued.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W.
3. R2 = Select a value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M12, N1-N12, P1-P12, Q1-Q12, R1, R2, S1-S6, T1, T2, U1-U6, V1-V6, W1-W12, and X1-X12. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions M13-M24, N13-N24, P13-P24, Q13-Q24, R3, R4, S7-S12, T3, T4, and U7-U12.

* FIGURE 11 EBL logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	1J	Input current, high level	2	Open	3,4,5,8,9,11,12,15, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	0	40	µA	12
J2	1K	"	4	5	2,3,8,9,11,12,15, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J3	2J	"	8	Open	2,3,4,5,9,11,12,15, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J4	2K	"	11	12	2,3,4,5,8,9,15,16, 17,18,21,22,23,24, 27,28,29,30,31,34, 35,36,37,38	"	"	"	"
J5	3J	"	15	Open	2,3,4,5,8,9,11,12, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J6	3K	"	17	18	2,3,4,5,8,9,11,12, 15,16,21,22,23,24, 27,28,29,30,31,34, 35,36,37,38	"	"	"	"
J7	4J	"	21	Open	2,3,4,5,8,9,11,12, 15,16,17,18,22,23, 24,28,29,30,31,34, 35,36,37,38	"	"	"	"
J8	4K	"	23	24	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 28,29,30,31,34,35, 36,37,38	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J9	5J	Input current, high level	28	Open	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,29,30,31, 35,36,37,38	0	40	µA	12
J10	5K	"	30	31	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,35, 36,37,38	"	"	"	"
J11	6J	"	35	Open	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,36,37,38	"	"	"	"
J12	6K	"	37	38	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,35,36	"	"	"	"
J13	1C	"	3	Open	2,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,35,36,37,38	-850	80	"	"
J14	2C	"	9	"	2,3,4,5,8,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,35,36,37,38	"	"	"	"
J15	3C	"	16	"	2,3,4,5,8,9,11,12, 15,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J16	4C	"	22	"	2,3,4,5,8,9,11,12, 15,16,17,18,21,23, 24,28,29,30,31,34, 35,36,37,38	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

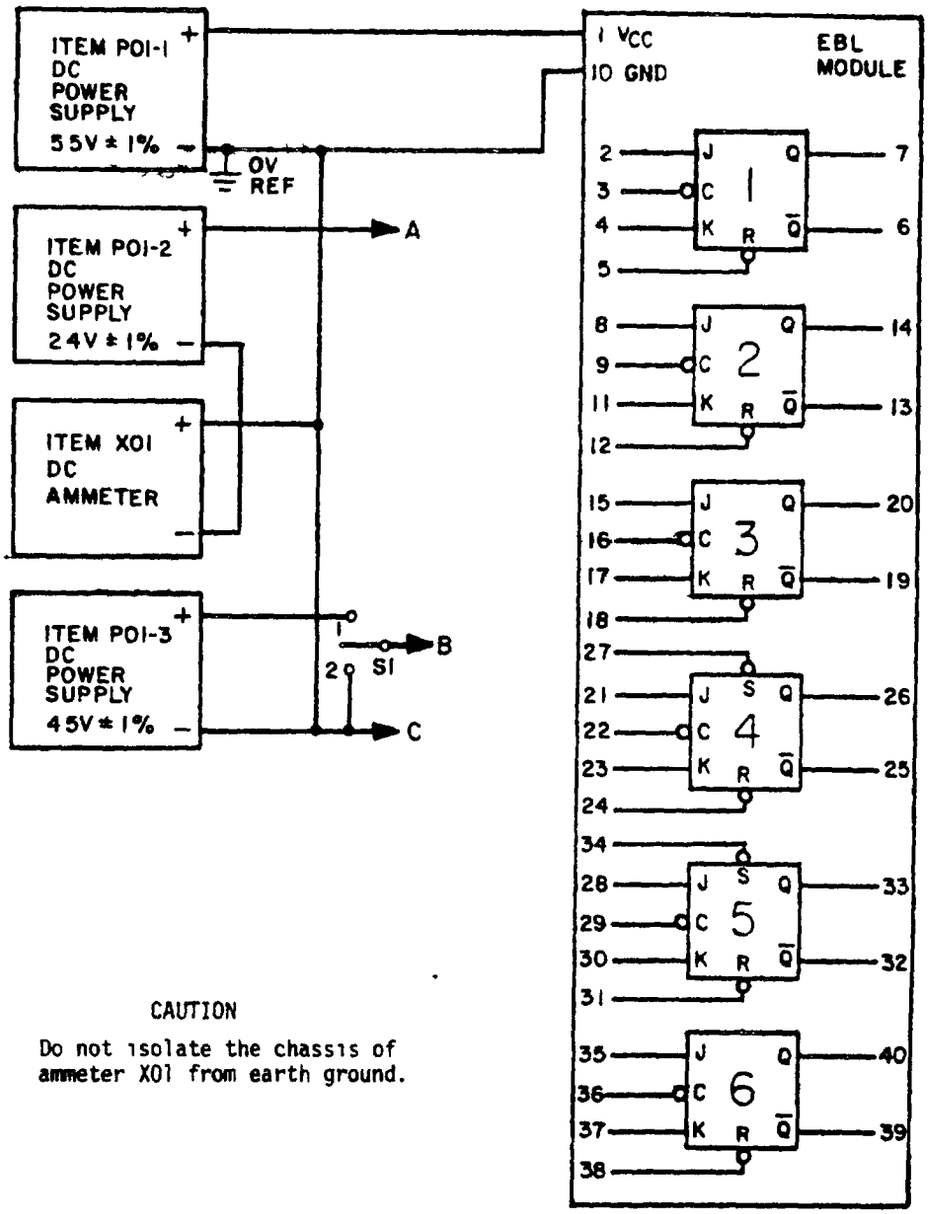
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J17	5C	Input current, high level	29	Open	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,30,31, 35,36,37,38	-850	80	µA	12
J18	6C	"	36	"	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,35,37,38	-850	"	"	"
J19	1R	"	5	"	2,3,4,8,9,11,12,15, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	0	"	"	"
J20	2R	"	12	"	2,3,4,5,8,9,11,15, 16,17,18,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J21	3R	"	18	"	2,3,4,5,8,9,11,12, 15,16,17,21,22,23, 24,27,28,29,30,31, 34,35,36,37,38	"	"	"	"
J22	4R	"	24	"	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,28,29,30,31,34, 35,36,37,38	"	"	"	"
J23	5R	"	31	"	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 35,36,37,38	"	"	"	"
J24	4S	"	27	"	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,28,29,30,31, 34,35,36,37,38	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J25	5S Input current, high level	34	Open	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,35,36,37,38	-210	-320	µA	12
J26	6R Input current, high level	38	Open	2,3,4,5,8,9,11,12, 15,16,17,18,21,22, 23,24,27,28,29,30, 31,34,35,36,37	-540	-690	µA	12

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CAUTION

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES.
1. S1 = SPDT switch.
2. Place switch S1 in position 2 for a minimum of 100 ns, then place switch S1 in position 1 and make measurement.

FIGURE 12. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	1J	Input current, low level	3,5	Open	2	-0.10	-1.73	mA	13
K2	1K	"	3,5	"	4	"	"	"	"
K3	2J	"	9,12	"	8	"	"	"	"
K4	2K	"	9,12	"	11	"	"	"	"
K5	3J	"	16,18	"	15	"	"	"	"
K6	3K	"	16,18	"	17	"	"	"	"
K7	4J	"	22,24	"	21	"	"	"	"
K8	4K	"	22,24	"	23	"	"	"	"
K9	5J	"	29,31	"	28	"	"	"	"
K10	5K	"	29,31	"	30	"	"	"	"
K11	6J	"	36	"	35	"	"	"	"
K12	6K	"	36	"	37	"	"	"	"
K13	1C	"	2,4	5	3	-0.20	-6.92	"	"
K14	1R	"	2,3	Open	5	"	"	"	"
K15	2C	"	8,11	12	9	"	"	"	"
K16	2R	"	8,9	Open	12	"	"	"	"
K17	3C	"	15,17	18	16	"	"	"	"
K18	3R	"	15,16	Open	18	"	"	"	"
K19	4C	"	21,23	24	22	"	"	"	"
K20	4R	"	21,22	Open	24	"	"	"	"

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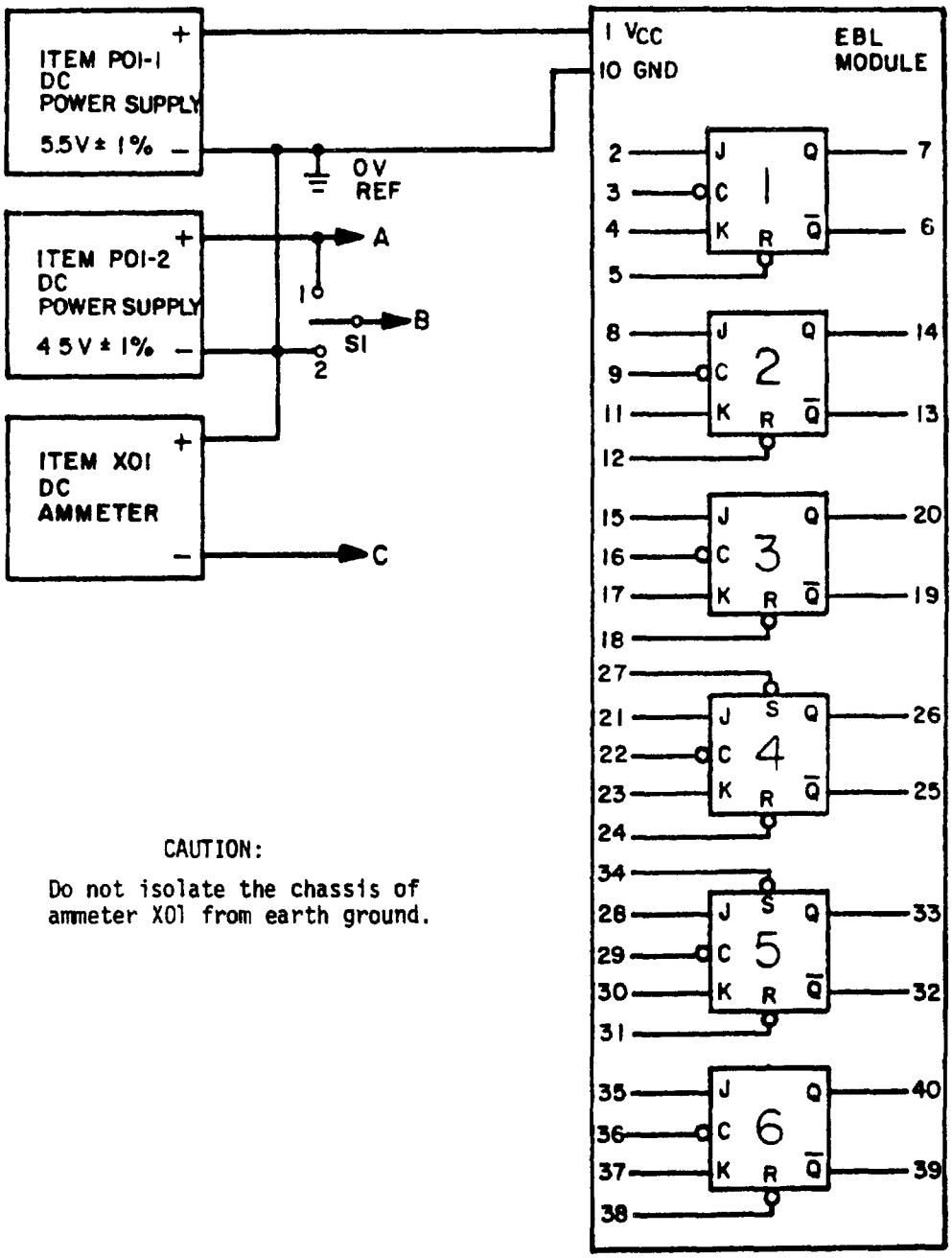
TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K21	4S	Input current, low level	22,23	Open	27	-0.20	-6.92	mA	13
K22	5C	"	28,30	31	29	"	"	"	"
K23	5R	"	28,29	Open	31	"	"	"	"
K24	6C	"	35,37	38	36	"	"	"	"
K25	5S	"	29,30	Open	34	-0.6	-7.5	"	"
K26	6R	"	35,36	Open	38	-1.20	-8.15	"	"

TABLE XII. Power supply current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Power supply current	27	Open	2,3,4,5,8,9, 11,12,15,16, 17,18,21,22, 23,24,28,29, 30,31,35,36, 37,38	40	120	mA	14
L2	Power supply current	2,5,8,12,15, 18,21,24,28, 31,35	3,9,16,36	4,11,17,23, 30,22,27,29, 34,37	40	120	mA	14

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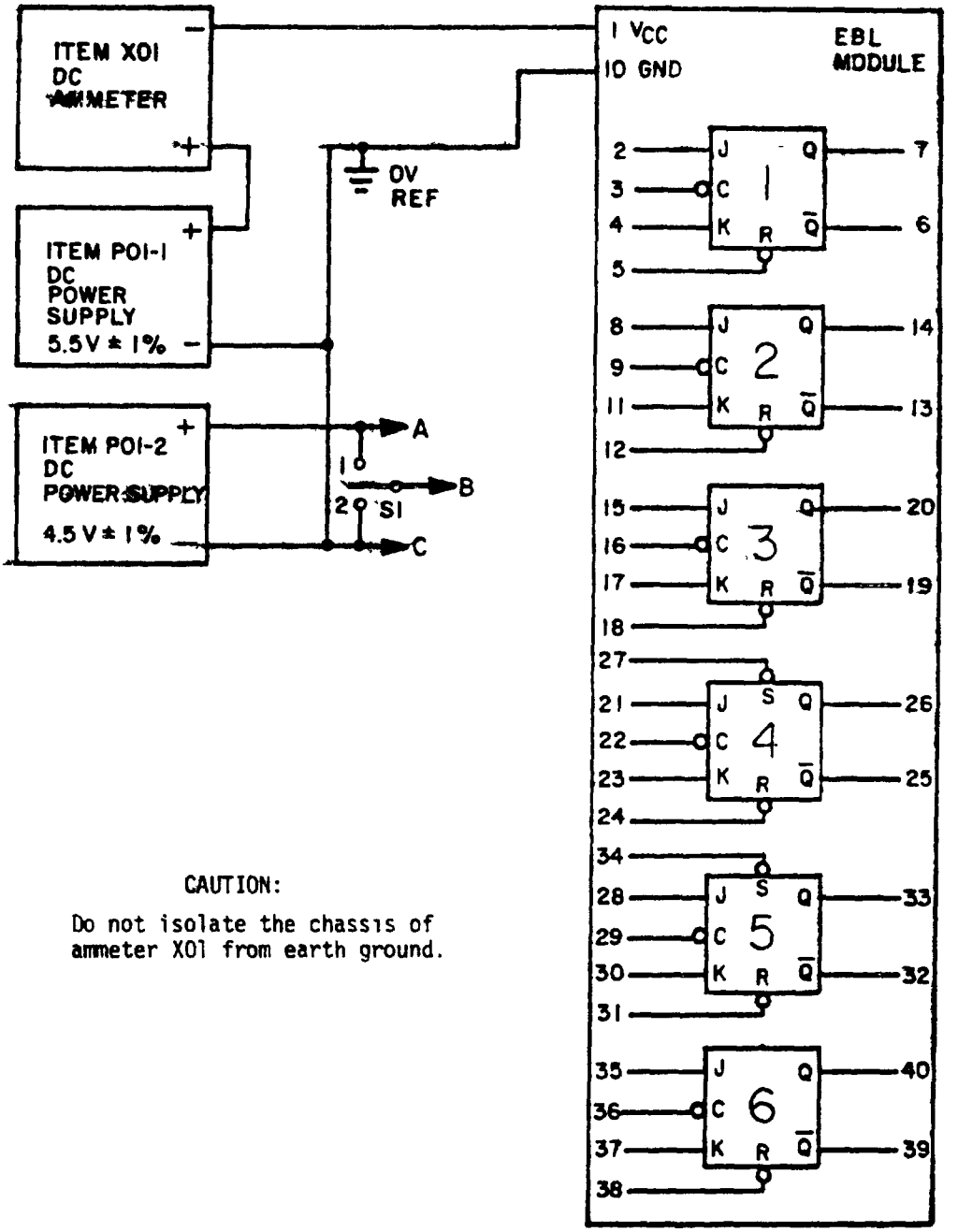
CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. S1 = SPDT switch.
2. Place switch S1 in position 2 for a minimum of 100 ns, then place S1 in position 1 and make measurement.

FIGURE 13. Input current (low-level) test circuit for table XI.

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. Place switch S1 in position 1 for a minimum of 100 ns, then place S1 in position 2 and make measurement.

FIGURE 14. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M1	1Q	Transition time (t_{TLH}), C_L = 50 pF	2,3,4	5	7	10	45	ns	15,16
M2	$1\overline{Q}$	"	2,3,4	5	6	"	"	"	"
M3	2Q	"	8,9,11	12	14	"	"	"	"
M4	$2\overline{Q}$	"	8,9,11	12	13	"	"	"	"
M5	3Q	"	15,16,17	18	20	"	"	"	"
M6	$3\overline{Q}$	"	15,16,17	18	19	"	"	"	"
M7	4Q	"	21,22,23	24	26	"	"	"	"
M8	$4\overline{Q}$	"	21,22,23	24	25	"	"	"	"
M9	5Q	"	28,29,30	31	33	"	"	"	"
M10	$5\overline{Q}$	"	28,29,30	31	32	"	"	"	"
M11	6Q	"	35,36,37	38	40	"	"	"	"
M12	$6\overline{Q}$	"	35,36,37	38	39	"	"	"	"
M13	1Q	Transition time (t_{TLH}), C_L = 150 pF	2,3,4	5	7	"	55	"	"
M14	$1\overline{Q}$	"	2,3,4	5	6	"	"	"	"
M15	2Q	"	8,9,11	12	14	"	"	"	"
M16	$2\overline{Q}$	"	8,9,11	12	13	"	"	"	"
M17	3Q	"	15,16,17	18	20	"	"	"	"
M18	$3\overline{Q}$	"	15,16,17	18	19	"	"	"	"
M19	4Q	"	21,22,23	24	26	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M20	4$\overline{Q}$	Transition time (t_{TLH}), C_L = 150 pF	21,22,23	24	25	10	55	ns	15,16
M21	5Q	"	28,29,30	31	33	"	"	"	"
M22	5$\overline{Q}$	"	28,29,30	31	32	"	"	"	"
M23	6Q	"	35,36,37	38	40	"	"	"	"
M24	6$\overline{Q}$	"	35,36,37	38	39	"	"	"	"
N1	1Q	Transition time (t_{THL}), C_L = 50 pF	2,3,4	5	7	5	25	"	"
N2	1$\overline{Q}$	"	2,3,4	5	6	"	"	"	"
N3	2Q	"	8,9,11	12	14	"	"	"	"
N4	2$\overline{Q}$	"	8,9,11	12	13	"	"	"	"
N5	3Q	"	15,16,17	18	20	"	"	"	"
N6	3$\overline{Q}$	"	15,16,17	18	19	"	"	"	"
N7	4Q	"	21,22,23	24	26	"	"	"	"
N8	4$\overline{Q}$	"	21,22,23	24	25	"	"	"	"
N9	5Q	"	28,29,30	31	33	"	"	"	"
N10	5$\overline{Q}$	"	28,29,30	31	32	"	"	"	"
N11	6Q	"	35,36,37	38	40	"	"	"	"
N12	6$\overline{Q}$	"	35,36,37	38	39	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N13	1Q	Transition time (t_{THL}), C_L = 150 pF	2,3,4	5	7	5	35	ns	15,16
N14	$1\overline{Q}$	"	2,3,4	5	6	"	"	"	"
N15	2Q	"	8,9,11	12	14	"	"	"	"
N16	$2\overline{Q}$	"	8,9,11	12	13	"	"	"	"
N17	3Q	"	15,16,17	18	20	"	"	"	"
N18	$3\overline{Q}$	"	15,16,17	18	19	"	"	"	"
N19	4Q	"	21,22,23	24	26	"	"	"	"
N20	$4\overline{Q}$	"	21,22,23	24	25	"	"	"	"
N21	5Q	"	28,29,30	31	33	"	"	"	"
N22	$5\overline{Q}$	"	28,29,30	31	32	"	"	"	"
N23	6Q	"	35,36,37	38	40	"	"	"	"
N24	$6\overline{Q}$	"	35,36,37	38	39	"	"	"	"
P1	1C to 1Q	Propagation delay time (t_{PLH}), C_L = 50 pF	2,3,4	5	7	10	30	"	"
P2	1C to $1\overline{Q}$	"	2,3,4	5	6	"	"	"	"
P3	2C to 2Q	"	8,9,11	12	14	"	"	"	"
P4	2C to $2\overline{Q}$	"	8,9,11	12	13	"	"	"	"
P5	3C to 3Q	"	15,16,17	18	20	"	"	"	"
P6	3C to $3\overline{Q}$	"	15,16,17	18	19	"	"	"	"
P7	4C to 4Q	"	21,22,23	24	26	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P8	4C to 4$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 50 pF	21,22,23	24	25	10	30	ns	15,16
P9	5C to 5Q	"	28,29,30	31	33	"	"	"	"
P10	5C to 5$\overline{Q}$	"	28,29,30	31	32	"	"	"	"
P11	6C to 6Q	"	35,36,37	38	40	"	"	"	"
P12	6C to 6$\overline{Q}$	"	35,36,37	38	39	"	"	"	"
P13	1C to 1Q	Propagation delay time (t_{PLH}), C_L = 150 pF	2,3,4	5	7	"	39	"	"
P14	1C to 1$\overline{Q}$		2,3,4	5	6	"	"	"	"
P15	2C to 2Q	"	8,9,11	12	14	"	"	"	"
P16	2C to 2$\overline{Q}$	"	8,9,11	12	13	"	"	"	"
P17	3C to 3Q	"	15,16,17	18	20	"	"	"	"
P18	3C to 3$\overline{Q}$	"	15,16,17	18	19	"	"	"	"
P19	4C to 4Q	"	21,22,23	24	26	"	"	"	"
P20	4C to 4$\overline{Q}$	"	21,22,23	24	25	"	"	"	"
P21	5C to 5Q	"	28,29,30	31	33	"	"	"	"
P22	5C to 5$\overline{Q}$	"	28,29,30	31	32	"	"	"	"
P23	6C to 6Q	"	35,36,37	38	40	"	"	"	"
P24	6C to 6$\overline{Q}$	"	35,36,37	38	39	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test - Continued.

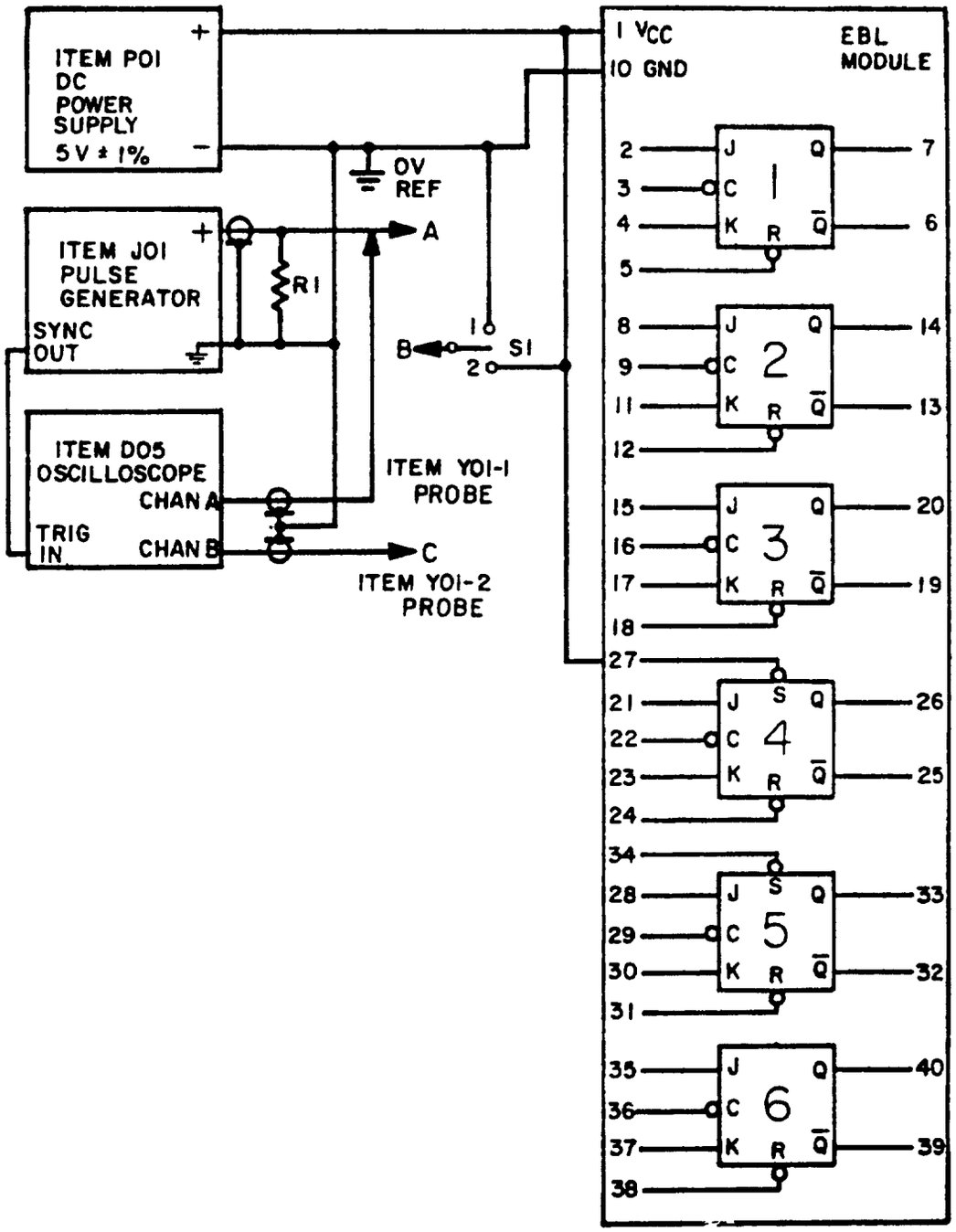
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	1C to 1Q	Propagation delay time (t_{PHL}), C_L = 50 pF	2,3,4	5	7	10	44	ns	15,16
Q2	1C to 1$\overline{Q}$	"	2,3,4	5	6	"	"	"	"
Q3	2C to 2Q	"	8,9,11	12	14	"	"	"	"
Q4	2C to 2$\overline{Q}$	"	8,9,11	12	13	"	"	"	"
Q5	3C to 3Q	"	15,16,17	18	20	"	"	"	"
Q6	3C to 3$\overline{Q}$	"	15,16,17	18	19	"	"	"	"
Q7	4C to 4Q	"	21,22,23	24	26	"	"	"	"
Q8	4C to 4$\overline{Q}$	"	21,22,23	24	25	"	"	"	"
Q9	5C to 5Q	"	28,29,30	31	33	"	"	"	"
Q10	5C to 5$\overline{Q}$	"	28,29,30	31	32	"	"	"	"
Q11	6C to 6Q	"	35,36,37	38	40	"	"	"	"
Q12	6C to 6$\overline{Q}$	"	35,36,37	38	39	"	"	"	"
Q13	1C to 1Q	Propagation delay time (t_{PHL}), C_L = 150 pF	2,3,4	5	7	"	54	"	"
Q14	1C to 1$\overline{Q}$	"	2,3,4	5	6	"	"	"	"
Q15	2C to 2Q	"	8,9,11	12	14	"	"	"	"
Q16	2C to 2$\overline{Q}$	"	8,9,11	12	13	"	"	"	"
Q17	3C to 3Q	"	15,16,17	18	20	"	"	"	"
Q18	3C to 3$\overline{Q}$	"	15,16,17	18	19	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q19	4C to 4Q	Propagation delay time (t_{PHL}), C_L = 150 pF	21,22,23	24	26	10	54	ns	15,16
Q20	4C to $4\overline{Q}$	"	21,22,23	24	25	"	"	"	"
Q21	5C to 5Q	"	28,29,30	31	33	"	"	"	"
Q22	5C to $5\overline{Q}$	"	28,29,30	31	32	"	"	"	"
Q23	6C to 6Q	"	35,36,37	38	40	"	"	"	"
Q24	6C to $6\overline{Q}$	"	35,36,37	38	39	"	"	"	"

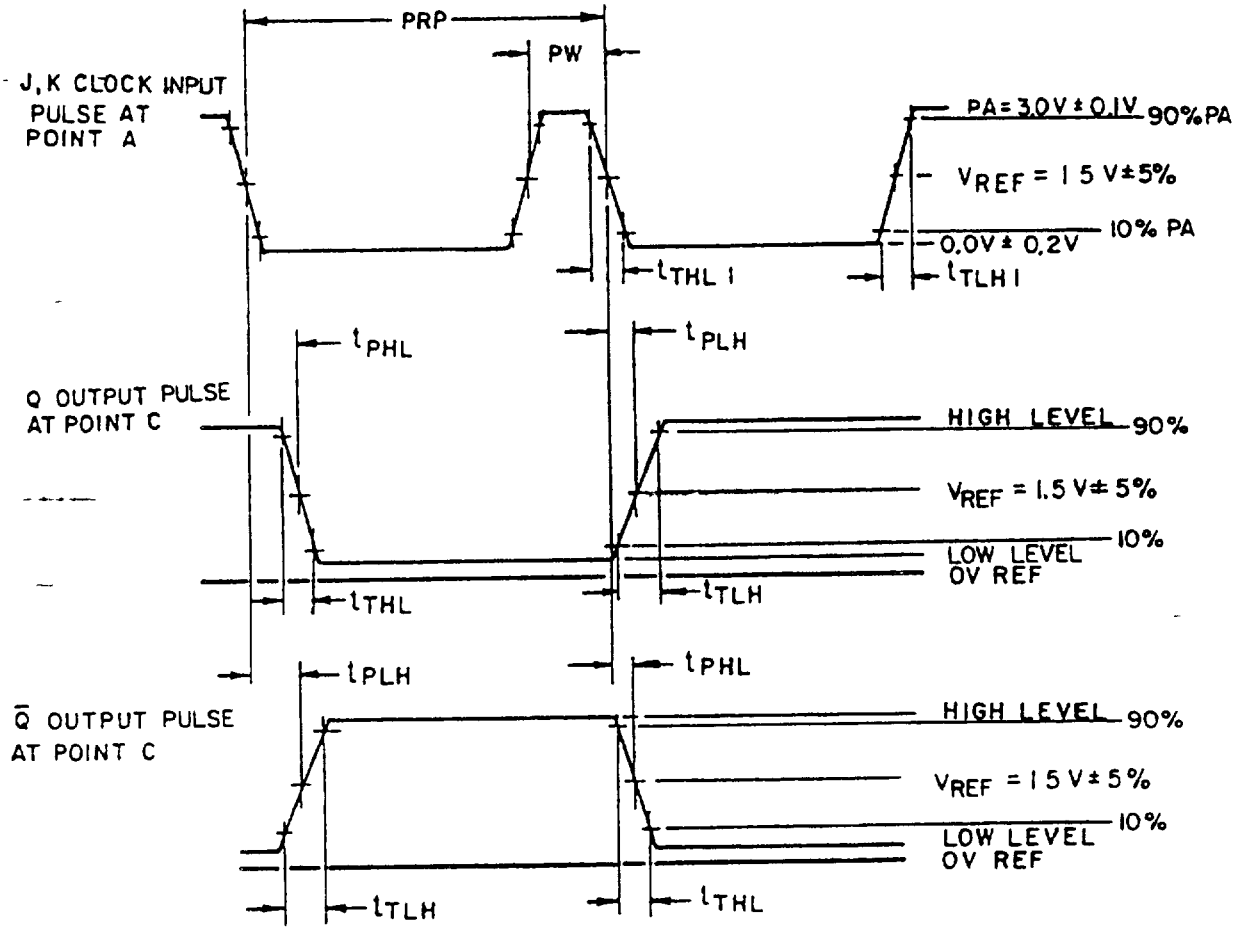
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. S1 = SPDT switch.
3. Load Q and $\overline{Q}$ outputswith load as specified on figure 11.
4. After making all other circuit connections per each test condition, place switch S1 in position 1 for a minimum of 100 ns, then place S1 in position 2 and make measurement.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test circuit for table XIII.

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NOTES

1. Input pulse parameters.
   PW = 21 ±1 ns
   PRP = 1,000 ±100 ns.
   Input t_{TLH} = t_{THL} = 10 ±2 ns.
2. Since the input pulse is connected to the J, K, and clock inputs, the requirements of a minimum input setup time equal to the clock pulse width and a minimum hold time of 0 ns are checked.
3. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
4. All voltage levels are referenced to ground.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum setup and hold time check) test circuit waveforms for table XIII.

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) (minimum reset and set pulse width check) test.

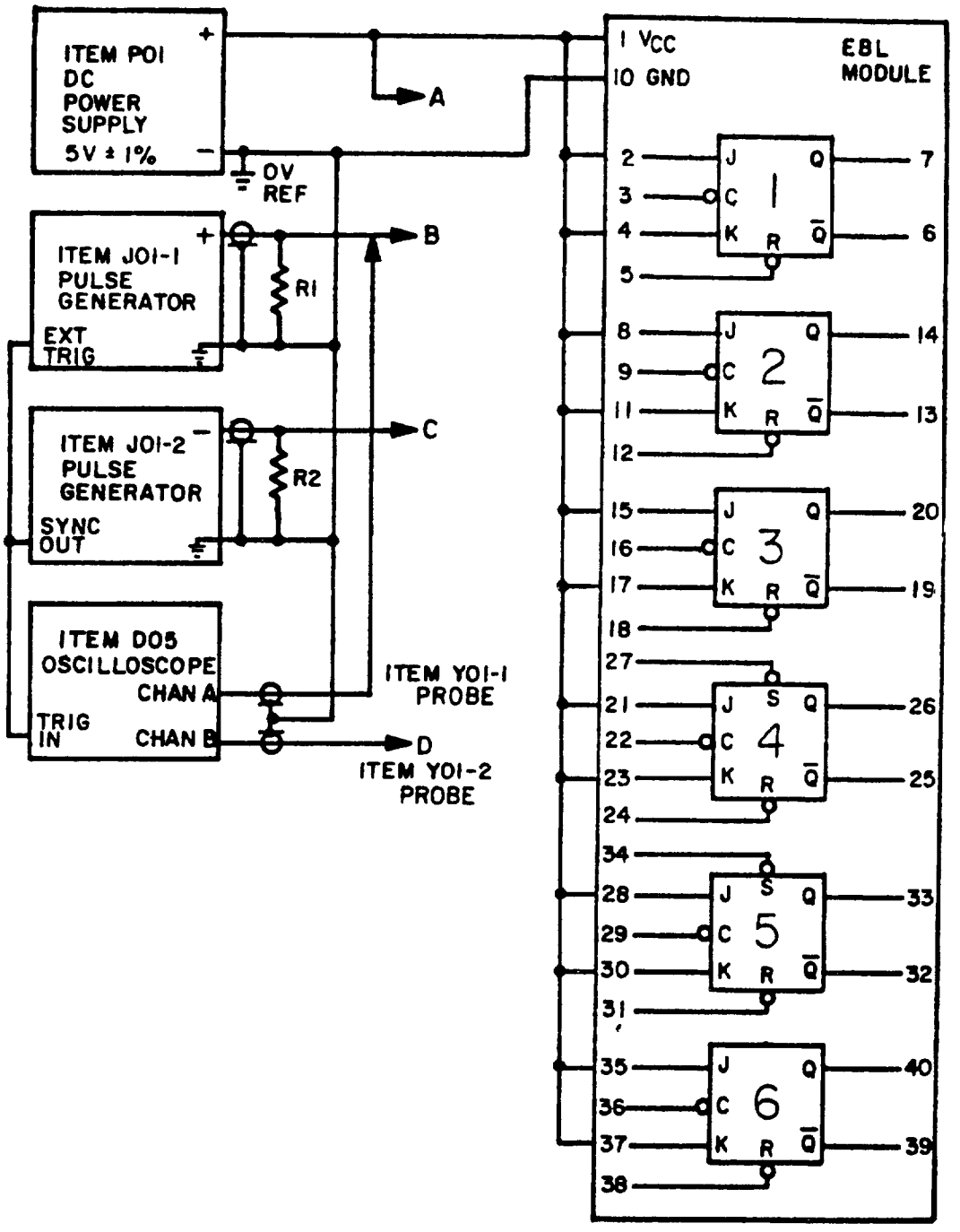
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	4S to 4Q Propagation delay time (t_{PLH}), C_L = 50 pF	24	22	27	26	5	29	ns	17,18
R2	5S to 5Q Propagation delay time (t_{PLH}), C_L = 50 pF	31	29	34	33	"	29	"	"
R3	4S to 4Q Propagation delay time (t_{PLH}), C_L = 150 pF	24	22	27	26	"	39	"	"
R4	5S to 5Q Propagation delay time (t_{PLH}), C_L = 150 pF	31	29	34	33	"	39	"	"
S1	1R to $1\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 50 pF	Open	3	5	6	"	29	"	"
S2	2R to $2\overline{Q}$ "	Open	9	12	13	"	"	"	"
S3	3R to $3\overline{Q}$ "	Open	16	18	19	"	"	"	"
S4	4R to $4\overline{Q}$ "	27	22	24	25	"	"	"	"
S5	5R to $5\overline{Q}$ "	Open	29	31	32	"	"	"	"
S6	6R to $6\overline{Q}$ "	Open	36	38	39	"	"	"	"
S7	1R to $1\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 150 pF	Open	3	5	6	"	39	"	"
S8	2R to $2\overline{Q}$ "	Open	9	12	13	"	"	"	"
S9	3R to $3\overline{Q}$ "	Open	16	18	19	"	"	"	"
S10	4R to $4\overline{Q}$ "	27	22	24	25	"	"	"	"
S11	5R to $5\overline{Q}$ "	Open	29	31	32	"	"	"	"
S12	6R to $6\overline{Q}$ "	Open	36	38	39	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) (minimum reset and set pulse width check) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	4S to $4\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF	24	22	27	25	10	44	ns	17,18
T2	5S to $5\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF	31	29	34	32	"	44	"	"
T3	4S to $4\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 150 pF	24	22	27	25	"	54	"	"
T4	5S to $5\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 150 pF	31	29	34	32	"	54	"	"
U1	1R to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF	Open	3	5	7	"	44	"	"
U2	2R to 2Q "	Open	9	12	14	"	"	"	"
U3	3R to 3Q "	Open	16	18	20	"	"	"	"
U4	4R to 4Q "	27	22	24	26	"	"	"	"
U5	5R to 5Q "	Open	29	31	33	"	"	"	"
U6	6R to 6Q "	Open	36	38	40	"	"	"	"
U7	1R to 1Q Propagation delay time (t_{PHL}), C_L = 150 pF	Open	3	5	7	"	54	"	"
U8	2R to 2Q "	Open	9	12	14	"	"	"	"
U9	3R to 3Q "	Open	16	18	20	"	"	"	"
U10	4R to 4Q "	27	22	24	26	"	"	"	"
U11	5R to 5Q "	Open	29	31	33	"	"	"	"
U12	6R to 6Q "	Open	36	38	40	"	"	"	"

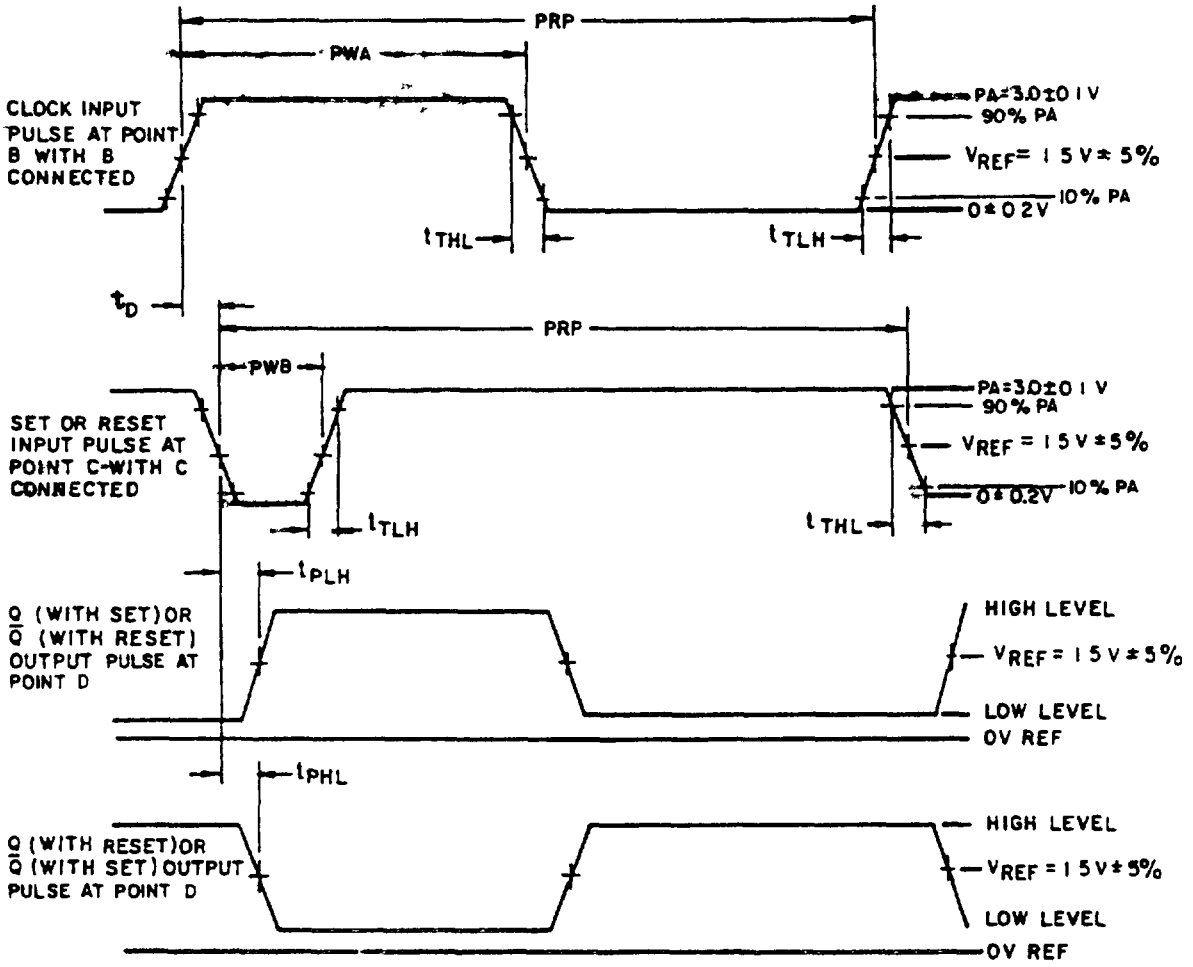
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) (minimum reset and set pulse width check) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   PWB = 26 ±1 ns (minimum set or reset pulse width).
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_D = 50 \pm 10$ ns.
2. Before making measurement, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

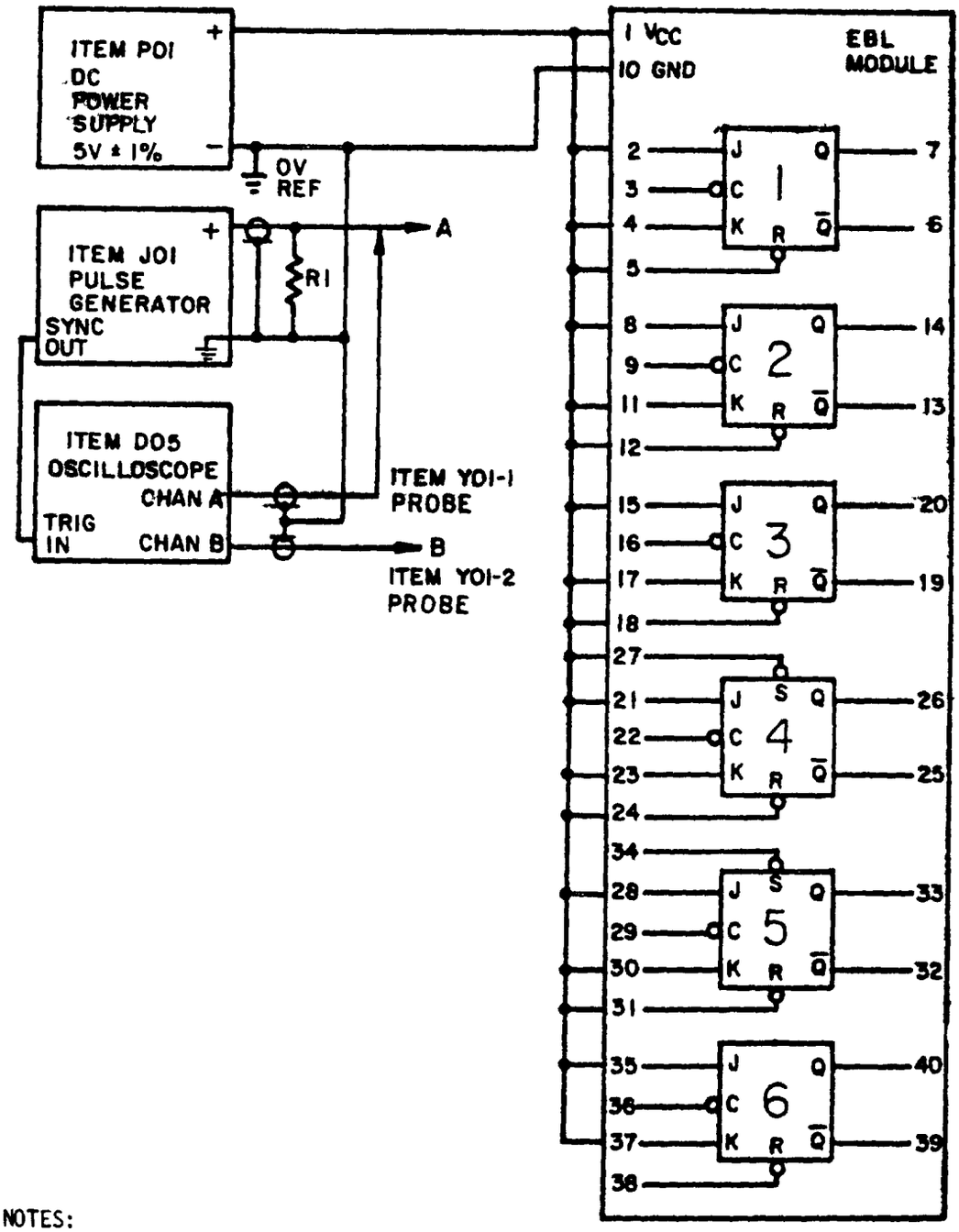
FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) (minimum reset and set pulse width check) test circuit waveforms for table XIV.

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TABLE XV. Maximum clock frequency check (minimum clock pulse width check) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	Maximum clock frequency check (1Q output period), C_L = 50 pF	3	7	200	212	ns	19,20
V2	Maximum clock frequency check (2Q output period), C_L = 50 pF	9	14	"	"	"	"
V3	Maximum clock frequency check (3Q output period), C_L = 50 pF	16	20	"	"	"	"
V4	Maximum clock frequency check (4Q output period), C_L = 50 pF	22	26	"	"	"	"
V5	Maximum clock frequency check (5Q output period), C_L = 50 pF	29	33	"	"	"	"
V6	Maximum clock frequency check (6Q output period), C_L = 50 pF	36	40	"	"	"	"

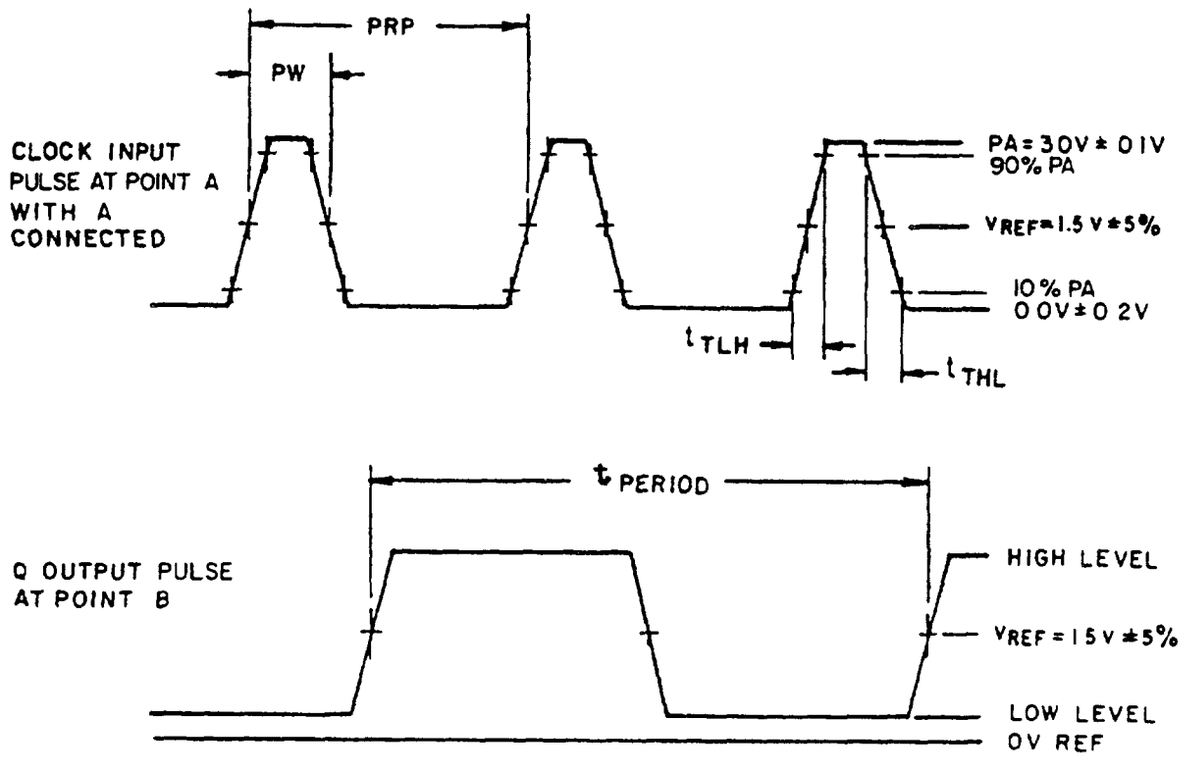
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 19. Maximum clock frequency check (minimum clock pulse width check) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PW = 21 ±1 ns (minimum clock pulse width).
   PRP = 103 ±3 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. Before making measurement, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

FIGURE 20. Maximum clock frequency check (minimum clock pulse width check) test circuit waveforms for table XV.

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TABLE XVI. Output crosstalk (high-level and low-level) test.

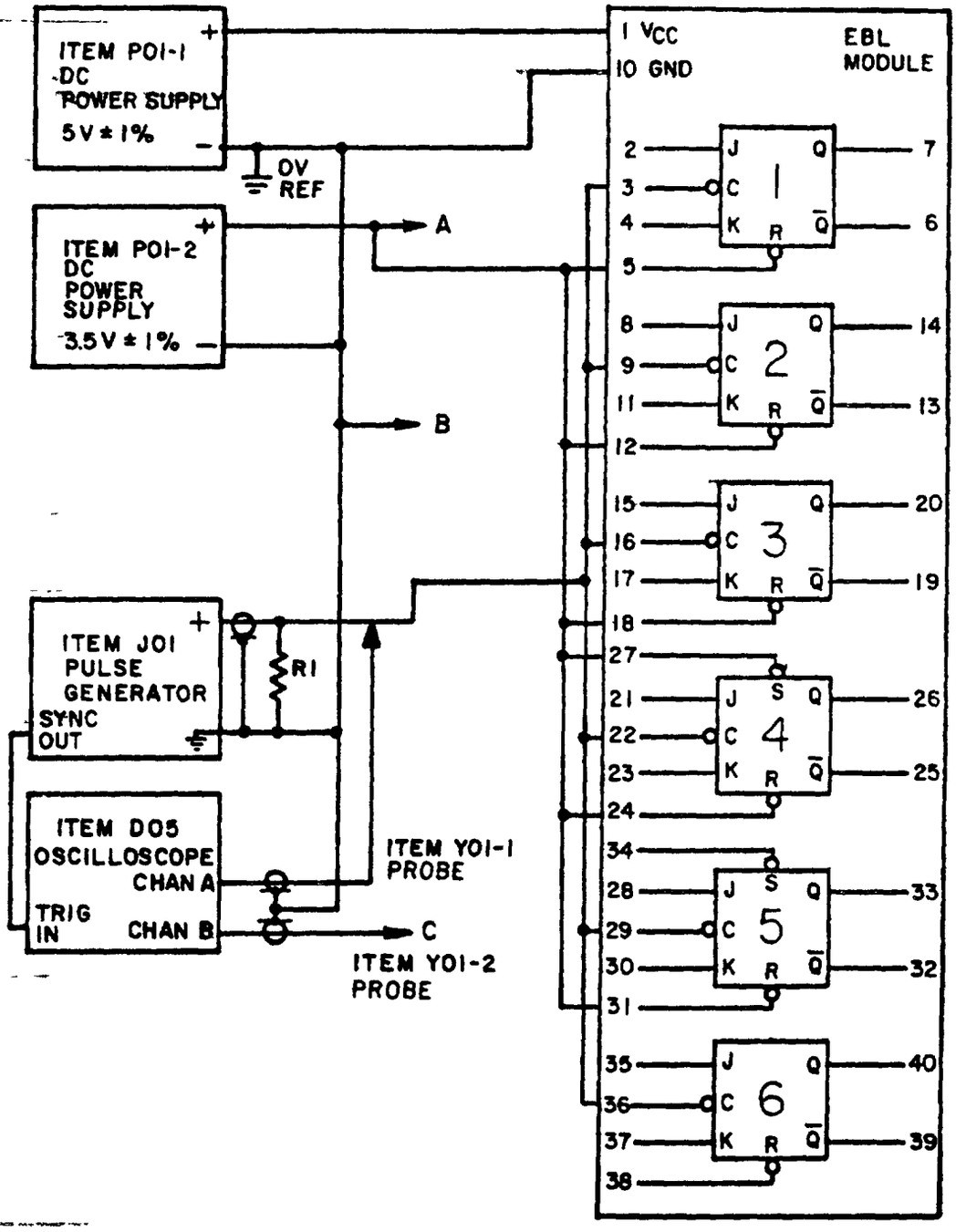
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1Q Crosstalk, high level	2,8,11,15,17,21,23,28, 30,35,37	4	7	0	400	mV	21,22
W2	$1\overline{Q}$ "	4,8,11,15,17,21,23,28, 30,35,37	2	6	"	"	"	"
W3	2Q "	2,4,8,15,17,21,23,28, 30,35,37	11	14	"	"	"	"
W4	$2\overline{Q}$ "	2,4,11,15,17,21,23,28, 30,35,37	8	13	"	"	"	"
W5	3Q "	2,4,8,11,15,21,23,28, 30,35,37	17	20	"	"	"	"
W6	$3\overline{Q}$ "	2,4,8,11,17,21,23,28, 30,35,37	15	19	"	"	"	"
W7	4Q "	2,4,8,11,15,17,21,28, 30,35,37	23	26	"	"	"	"
W8	$4\overline{Q}$ "	2,4,8,11,15,17,23,28, 30,35,37	21	25	"	"	"	"
W9	5Q "	2,4,8,11,15,17,21,23, 28,35,37	30	33	"	"	"	"
W10	$5\overline{Q}$ "	2,4,8,11,15,17,21,23, 30,35,37	28	32	"	"	"	"
W11	6Q "	2,4,8,11,15,17,21,23, 28,30,35	37	40	"	"	"	"
W12	$6\overline{Q}$ "	2,4,8,11,15,17,21,23, 28,30,37	35	39	"	"	"	"

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TABLE XVI. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1Q Crosstalk, low level	4,8,11,15,17,21,23,28, 30,35,37	2	7	0	400	mV	21,22
X2	$1\overline{Q}$ "	2,8,11,15,17,21,23,28, 30,35,37	4	6	"	"	"	"
X3	2Q "	2,4,11,15,17,21,23,28, 30,35,37	8	14	"	"	"	"
X4	$2\overline{Q}$ "	2,4,8,15,17,21,23,28, 30,35,37	11	13	"	"	"	"
X5	3Q "	2,4,8,11,17,21,23,28, 30,35,37	15	20	"	"	"	"
X6	$3\overline{Q}$ "	2,4,8,11,15,21,23,28, 30,35,37	17	19	"	"	"	"
X7	4Q "	2,4,8,11,15,17,23,28, 30,35,37	21	26	"	"	"	"
X8	$4\overline{Q}$ "	2,4,8,11,15,17,21,28, 30,35,37	23	25	"	"	"	"
X9	5Q "	2,4,8,11,15,17,21,23, 30,35,37	28	33	"	"	"	"
X10	$5\overline{Q}$ "	2,4,8,11,15,17,21,23, 28,35,37	30	32	"	"	"	"
X11	6Q "	2,4,8,11,15,17,21,23, 28,30,37	35	40	"	"	"	"
X12	$6\overline{Q}$ "	2,4,8,11,15,17,21,23, 28,30,35	37	39	"	"	"	"

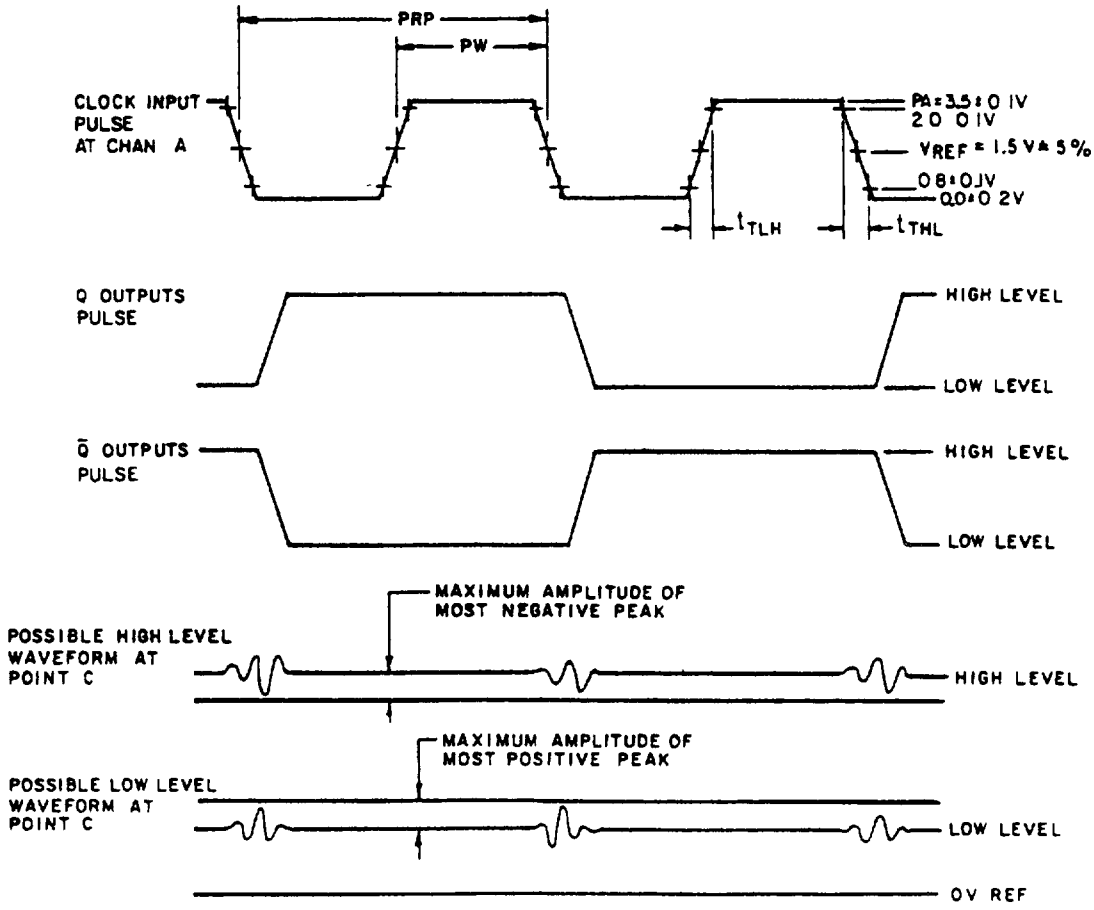
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 21. Output crosstalk (high-level and low-level) test circuit for table XVI.

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NOTE: Input pulse parameters.
PW = 500 ±50 ns
PRP = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin)

FIGURE 22. Output crosstalk (high-level and low-level) test circuit waveforms for table XVI.

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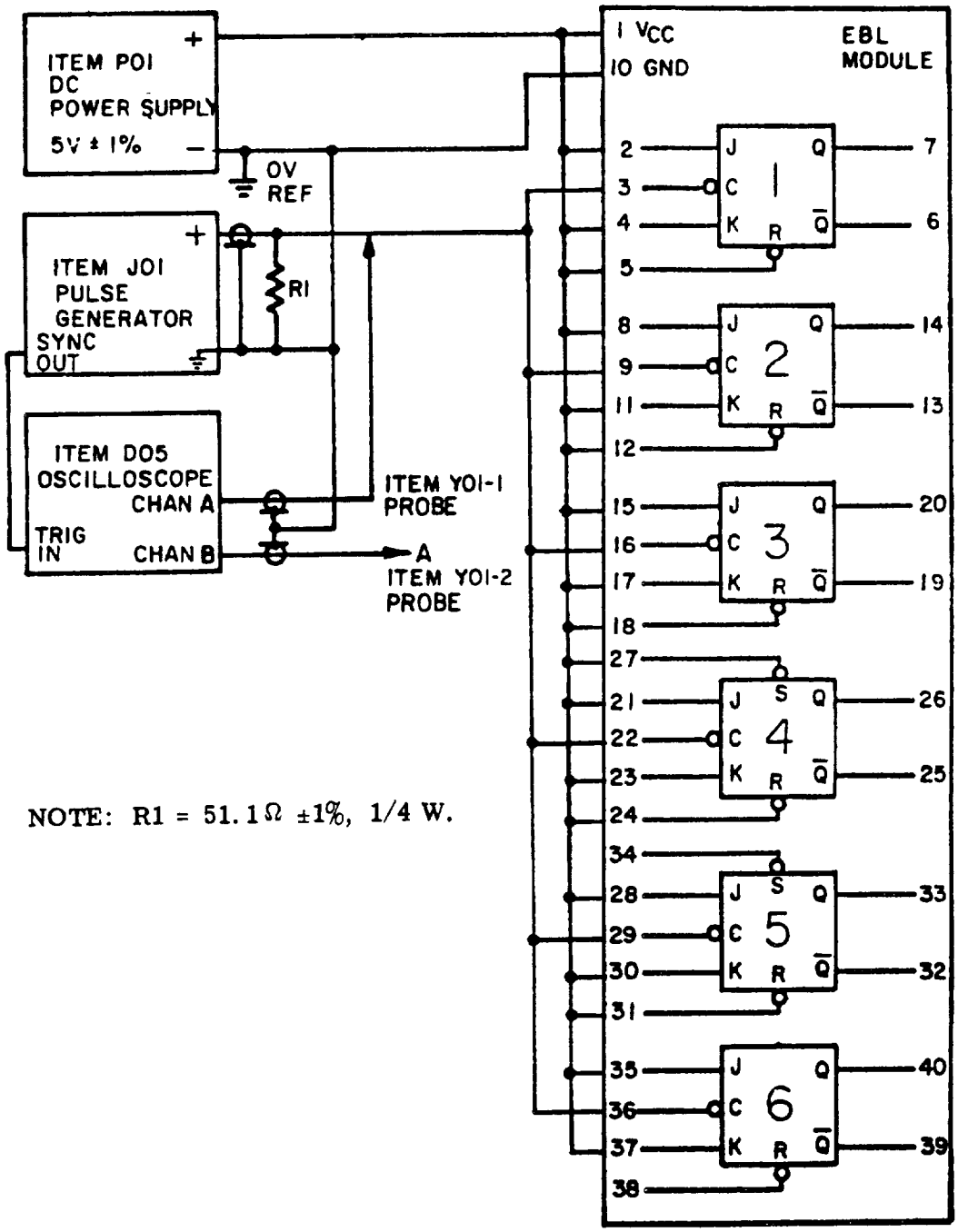
TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Y	Power supply transient amplitude	1	0	400	mV	23,24

TABLE XVIII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply filter capacitance	1.6	3.0	µF	25

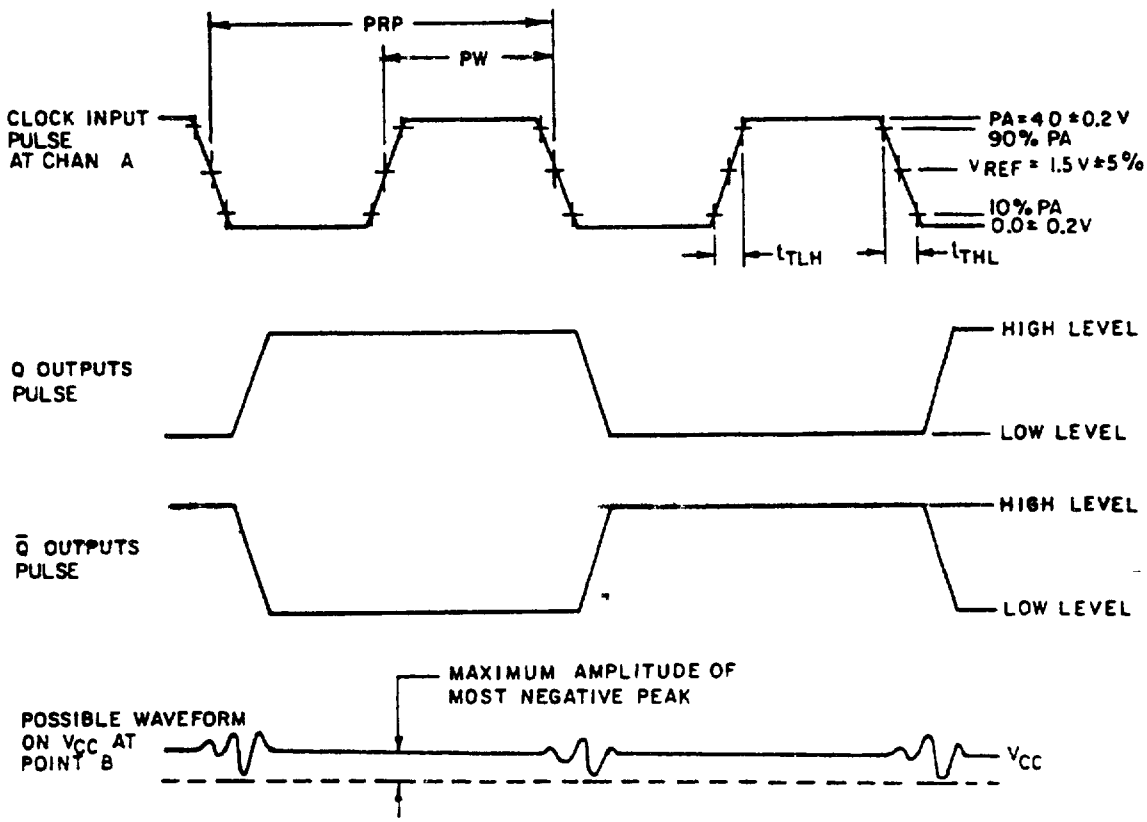
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVII.

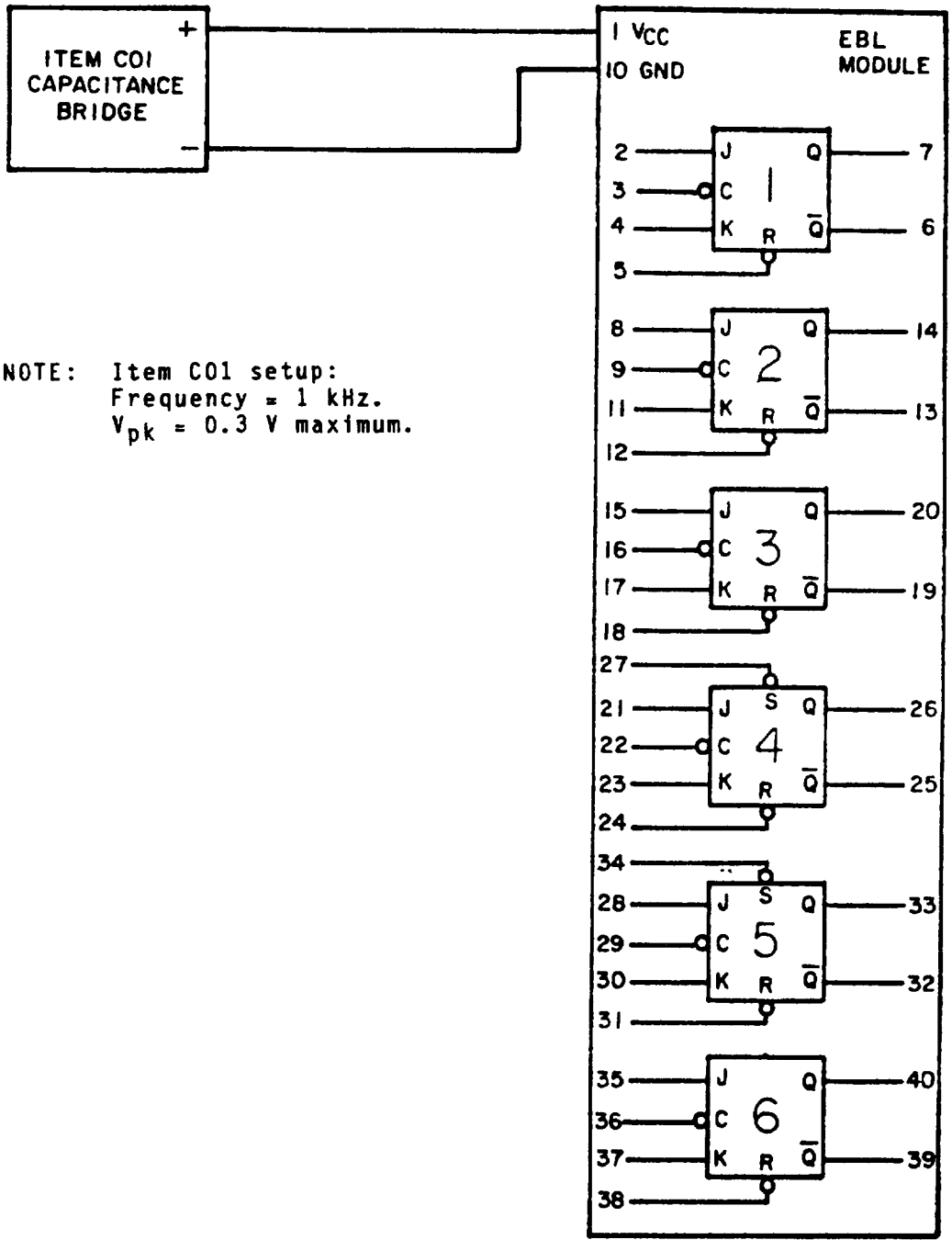
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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVII.

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NOTE: Item C01 setup:
Frequency = 1 kHz.
V_{pk} = 0.3 V maximum.

FIGURE 25. Power supply filter capacitance test circuit for table XVIII.

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TABLE XIX. Input capacitance test.

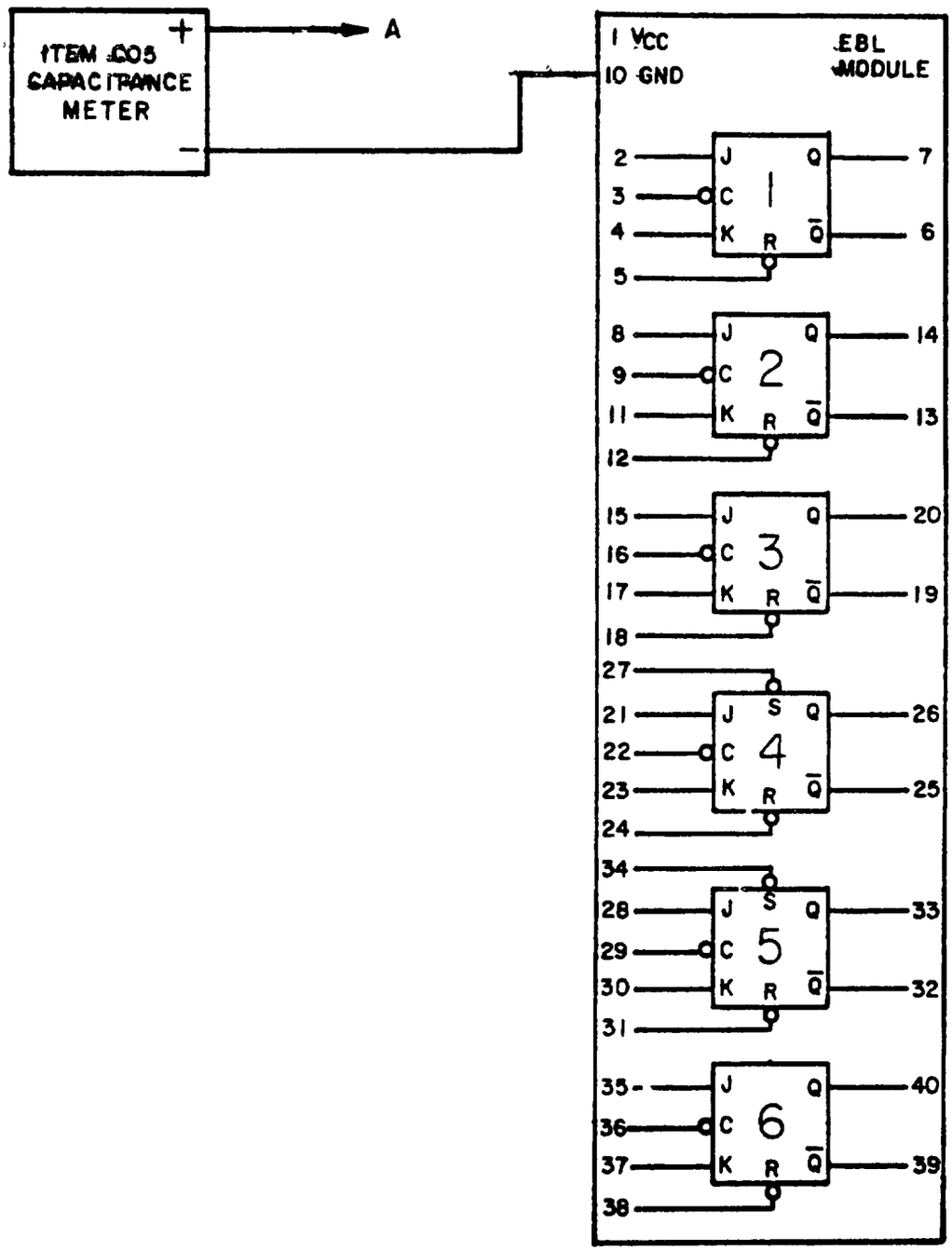
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
AA1	1J	Input capacitance	2	2	15	pF	26
AA2	1K	"	4	"	"	"	"
AA3	2J	"	8	"	"	"	"
AA4	2K	"	11	"	"	"	"
AA5	3J	"	15	"	"	"	"
AA6	3K	"	17	"	"	"	"
AA7	4J	"	21	"	"	"	"
AA8	4K	"	23	"	"	"	"
AA9	5J	"	28	"	"	"	"
AA10	5K	"	30	"	"	"	"
AA11	6J	"	35	"	"	"	"
AA12	6K	"	37	"	"	"	"
AA13	1C	"	3	"	24	"	"
AA14	1R	"	5	"	"	"	"
AA15	2C	"	9	"	"	"	"
AA16	2R	"	12	"	"	"	"
AA17	3C	"	16	"	"	"	"
AA18	3R	"	18	"	"	"	"
AA19	4C	"	22	"	"	"	"

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TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
AA20	4S	Input capacitance	27	2	24	pF	26
AA21	4R	"	24	"	"	"	"
AA22	5C	"	29	"	"	"	"
AA23	5S	"	34	"	"	"	"
AA24	5R	"	31	"	"	"	"
AA25	6C	"	36	"	"	"	"
AA26	6R	"	38	"	"	"	"

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the reading.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 26. Input capacitance test circuit for table XIX.

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

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